

Cash Management and Payment Choices Revisited

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Abstract

Exploiting transaction-level microdata from European payment diaries, I show that existing models of payment choice and cash management fail to capture relevant features of the data. I build a novel, dynamic model of cash management with a payment method choice and lumpy expenditures which rationalizes two key characteristics of payment behavior: i) consumers do not choose among cash and cards based on a threshold transaction size; ii) cashless methods are sometimes employed even if cash on hand is sufficient. I show that, as long as cashless payments are *cheaper* than ATM withdrawals, agents may find it optimal to use their cards even when they have sufficient cash on hand, in order to keep cash holdings close to their optimal levels. The model's payment choice policies can rationalize observed choices that earlier studies could not account for. After calibrating the model to qualitatively match cross-country heterogeneity in payment and cash management patterns, I discuss possible extensions which would allow for a quantitative analysis.

Keywords: *payment choice, cash management, cashless payments, inventory models.*

JEL Codes: *E41, E42, D14.*

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1 Introduction and overview

In this paper, I exploit unique, transaction-level data for the Euro Area and I present a novel conceptual framework to shed new light on the determinants of payment choices and cash management. Drawing upon detailed payment diaries provided in the ECB SUCH (2016) and SPACE (2019) studies, which enable me to observe consumers' *payment choice sets*, I show that existing theoretical models fail to capture some relevant dimensions of real-world payment behavior. In particular, I present evidence indicating that *transaction size policies* à la Whitesell (1989) and *cash burns policies* à la Alvarez and Lippi (2017) are not consistent with actual choices at POS. Combining payment diaries and answers to survey questionnaires, I show that i) it is crucial to jointly study cash management and payment choices; ii) cash/cashless decisions are strongly influenced both by cash holdings m and by the size s of transactions faced; iii) supply-side factors such as non-universal cashless acceptance shape cash management and payment choice policies by households.

Then, I present a novel, dynamic model of payment choice and cash management which includes these elements, combining features of Whitesell (1989) (lumpy, heterogeneous transaction sizes) and Alvarez and Lippi (2017) (payment choices that depend on cash holdings) in a unified framework. The model is built on a discrete-time version of the standard Baumol-Tobin cash management framework (Baumol (1952), Tobin (1956)) with some modifications. First, as in earlier models of payment choice (Whitesell (1989), Alvarez and Lippi (2017)) agents are endowed with a cashless payment instrument, whose usage entails a fixed cost κ ¹. Second, agents face an exogenous, stochastic expenditure stream, in lieu of a constant, deterministic one. In any period, they face a random transaction size s drawn from a known distribution: before observing s , depending on current cash holdings m , they can decide whether to withdraw money by paying a fixed withdrawal cost η ². After s realizes, agents must decide whether to use cash or cashless, depending on current cash holdings and on the size of the transaction faced. To perform a cash payment, agents need to have enough cash on hand ($m \geq s$): as in all cash management models, holding cash entails an opportunity cost. Moreover, the possibility to use cards is not always granted, as cashless acceptance is non-universal. The uncertainty induced by the stochastic expenditure flow and by imperfect card acceptance induce a precautionary demand for cash holdings. The presence of lumpy transactions generates realistic withdrawal behavior (positive cash holdings at withdrawal, dispersion in withdrawal sizes) without the need to allow

¹This fixed cost can represent a monetary fee which is paid when using the card. It also captures other kinds of costs perceived by consumers when using cashless methods: privacy costs (as the transaction can be perceived as not being anonymous anymore) and time costs (as the literature as shown that cash payments are settled more quickly, see Klee (2008))

²This fixed cost represents both fees paid when withdrawing cash from ATMs and time costs associated with going to the bank (the famous *shoeleather costs*, as they are labeled in the monetary economics literature).

for free withdrawal opportunities, as done in [Alvarez and Lippi \(2017\)](#).

The presence of a two-dimensional state space and of stochastic, lumpy transaction sizes makes the model too complicated to obtain analytical, pencil-and-paper solutions. However, under some assumptions (which can be easily verified after solving the model numerically) I can derive results on optimal cash management and payment choice policies. The core result of this paper is presented in [Proposition 2](#), which derives features of the payment choice policy function. My model is the first to rationalize the usage of cashless payments even by consumers who have enough cash on hand to settle a transaction. I obtain a key relationship between the cost of a withdrawal and that of a cashless payment (relative to a cash transaction) which needs to hold in order to induce voluntary card usage. If paying cashless is more costly than paying by cash, but cheaper than withdrawing money from ATMs, agents may find it optimal to use their cards even when they have sufficient cash on hand, in order to keep cash holdings close to their optimal levels. In other words, in some situations cashless payments are employed as a *cash management tool*, if they are a cheaper way to control cash balances than ATM withdrawals. I show that the model's decision rules can replicate features of payment behavior which were not explained by previous theories, rationalize a much larger share of transactions in SUCH and SPACE data and solve an apparent *payment choice puzzle* observed in the data (namely, the fact that agents who reportedly prefer cash sometimes use their cards even when they are not forced to do so).

Then, I show that a calibrated version of the model can qualitatively replicate differences in payment and cash management patterns across countries. At the end of the paper, I discuss possible extensions of the model that would make it suitable for a quantitative study of payment choices and for identifying parameters using tools from the dynamic discrete choice literature.

1.1 Related literature and contribution

In recent years, innovations in financial markets and information technologies widened the set of payment methods available for individuals. Already widespread tools such as credit and debit cards became more attractive, as the birth of contactless technologies markedly improved their efficiency. As a result, the share of cashless payments has been increasing all over the world. As cash/cashless choices became of wider interest for economists, a vast literature studying the determinants of consumers' payment choices has grown.

A large part of this literature is empirical, and it made use of payment diaries or stores' transaction data. Some of the findings of this literature are extremely relevant for models of payment choice. Exploiting grocery store data, [Klee \(2008\)](#) examines the relationship between transaction sizes s and payment choices, finding that cash is mainly used for small-sized purchases, while cashless payments are prevalent when the value

of the sale increases. Similar results are obtained by Wang and Wolman (2016) using two billion retail transactions. Looking at the association between cash holdings m and payment choices, several papers (Arango, Huynh, Fung, et al. (2012), Bouhdaoui and Bounie (2012), Huynh, Schmidt-Dengler, and Stix (2014), Bagnall et al. (2016)) find that the likelihood of cash usage increases with the level of cash holdings. Other studies (Arango, Huynh, and Sabetti (2015)) use survey data to show the relevance of the probability of merchant acceptance in shaping means of payments decisions by consumers. Huynh, Schmidt-Dengler, and Stix (2014) and Bagnall et al. (2016) observe an interplay between payment choices, card acceptance and cash management decisions, finding that in economies characterized by low card acceptance rates, people hold a precautionary buffer stock of cash, seemingly in order to avoid repeated withdrawals, as they are frequently forced to pay large cash sums. In this paper, I contribute to this empirical literature in three ways. First, I provide further evidence that payment choices have an impact on cash management policies. Second, I show that payment choices by households are influenced by cash management concerns, i.e., that agents decide whether to pay with cash and cashless taking into account the effect of their choice on future cash management costs. Third, I document that expected levels of card acceptance affect both cash management patterns and payment method decisions.

There are several papers that study theoretically means of payment choices, but only a few try to jointly model cash management and payment choices by households. The two most notable attempts are those by Whitesell (1989) and Alvarez and Lippi (2017), which will be discussed at length in the main text. Both papers rationalize well-established empirical regularities. Whitesell (1989) derives payment choice rules that depend on the size of the transaction, showing that it is optimal for agents to use cash when $s < \bar{s}$ and cashless methods otherwise (*transaction size threshold policies*). Alvarez and Lippi (2017) abstract from transaction size heterogeneity and derive payment choice rules that depend on cash holdings, showing that for small payment sizes cash usage is optimal whenever agents have enough (*cash burns policies*). In this paper, I contribute to this literature in two ways. First, I exploit high-quality microdata to show that a large share of actual transactions cannot be rationalized by the two policies discussed above. Second, I combining elements of both papers in a discrete-time Baumol-Tobin framework, allowing agents to decide on how to pay depending both on cash holdings m and on the size s of the transaction. Another novelty is the explicit incorporation of imperfect cashless acceptance. This work is an attempt to bridge empirical findings that show how payment choices are multi-dimensional and the existing theoretical literature, as advocated in the final Section of Wang and Wolman (2016).

1.2 Relevance

The joint study of the transaction demand for cash by households and of their payment method choices is policy relevant for a variety of reasons. A first possible application is using the model to estimate how consumers would be affected by different regulation schemes for debit/credit card interchange fees, a topic widely discussed in the IO literature ([Rochet and Tirole \(2002\)](#), [Li, McAndrews, and Wang \(2019\)](#)). Secondly, the availability of cashless payments has a first-order impact on the transaction demand for cash, with implications for the measurement of the welfare cost of inflation ([Alvarez and Lippi \(2009\)](#)). Finally, understanding the rationale behind payment choices by embedding them into models of cash management can be useful to evaluate quantitatively the welfare effects of proposed policies aimed at fighting illegal activities such as organized crime and tax evasion, implementable by banning cash ([Rogoff \(2017\)](#)) or discouraging its use ([Immordino and Russo \(2018\)](#)).

1.3 Structure of the paper

In [Section 2](#) I describe SUCH and SPACE data and I perform an empirical analysis of payment behavior in the Euro Area, highlighting some facts which cannot be rationalized by existing models of payment choice and documenting features that should be included in a more comprehensive theory. In [Section 3](#) I present my theoretical framework, I discuss properties of the solutions and I discuss how the payment choice policies that emerge from the model are a generalization of previously found ones. In [Section 4](#) I compare the predictions of the model with actual data on payment choices and cash management. In [Section 5](#) I calibrate the model to qualitatively replicate cross-country differences in payment and cash management behavior in the Euro Area. [Section 6](#) concludes.

2 Empirical Analysis

2.1 Data description

I start by providing an overview of the two data sources used in this paper. Throughout the analysis, I exploit microdata on payment choices and cash management behavior of individuals from 15 Euro Area countries, combining two datasets provided by the ECB jointly with two publications: the first one, released in 2016, is the Study on the Use of Cash by Households (SUCH from now on); the second one, released in 2020, is the Study on the Payment Attitudes of Consumers in the Euro Area (SPACE from now on)³. Each study contains a payment diary and a survey questionnaire: in the diary,

³For further information about the methodology and results of these two studies, see respectively [Esselink and Hernández \(2017\)](#) and [ECB \(2020\)](#).

TABLE 1: Summary statistics.

	Mean	Std. Dev.	Min	Max	N
<i>Payment choices</i>					
Payment size (€)	21.38	87.35	0.01	16,000.00	155565
Paid with cash	0.74	0.44	0.00	1.00	155565
Cashless accepted	0.74	0.44	0.00	1.00	152360
Cashless possible	0.68	0.47	0.00	1.00	129158
Cash possible	0.89	0.31	0.00	1.00	178472
Both methods possible	0.53	0.50	0.00	1.00	129169
Paid with cash Both possible	0.68	0.47	0.00	1.00	69013
<i>Cash management choices</i>					
Cash holdings (€)	68.47	152.86	0.00	9,990.92	178472
Withdrawn	0.03	0.18	0.00	1.00	90843
Withdrawal size (€)	91.93	161.11	0.45	3,600.00	4063
Adjusted	0.08	0.27	0.00	1.00	90843
Adjustment size (€)	75.03	145.01	0.05	3,600.00	6502

Note: Adjustments are defined as any increase in cash holdings between a payment and another; they also include receiving money from friends, or as cash income. Withdrawals are a particular type of adjustment: in these cases, agents either withdrew money from an ATM or got it from a bank teller. *Source: ECB SUCH (2016) and SPACE (2019) Data.*

participants recorded information on all their payments and adjustments of cash balances in a given day; within the questionnaire, they were asked a number of questions on their preferences about payment methods, on the set of cashless instruments they had access to and on their cash management behavior.

The main variables are summarized in Table 1. A key characteristic of these data is that one can combine the information provided to track down the time path of cash holdings and to elicit *payment choice sets* of consumers (whether cash or cashless or both could be used to carry out purchases) for each transaction during the day of analysis. This feature enables me to distinguish between *voluntary payments* and *forced* ones. I define a payment as *voluntary* when the agent has both options available: she holds sufficient cash, she has access to cashless payment method and the current store accepts cashless payments. Payment method choices that do not satisfy one of these conditions are *forced*, as in these situations the payment choice set collapses to a singleton ⁴.

The ability of observe cash holdings at each point in time creates a unique setup to

⁴Appendix A.1 contains details on the data cleaning procedure that enables me to derive the time path of cash holdings and therefore to elicit payment choice sets. In all the analysis, I abstract from imperfect cash acceptance. It is now quite common for stores to avoid accepting cash from clients. In SUCH data there is no information on this; in SPACE data agents were asked about cash being accepted, but the extent of recorded non-acceptance in the data is so tiny that I do not consider it explicitly.

study payment choices: in previous studies that analyzed payment behavior, payment choice sets are not appropriately taken into account because cash holdings are typically unobserved. [Klee \(2008\)](#) exploits grocery store data, thereby ruling out imperfect acceptance of cashless payments: however, her strategy does not differentiate between voluntary cashless payments (in which the agent also had the opportunity to pay with cash) and forced ones (in which the agent did not have enough cash to carry out the purchase). A similar problem is encountered by [Wakamori and Welte \(2017\)](#): while they do observe perceived payment acceptance by agents, they don't know how much cash do they have at each point in time. In both papers, the authors can *limit* the choice sets of agents, but do not *observe* the choice set in an exact way. The data used in this paper makes choice sets exactly observable: this allows me to understand if a choice is driven by preferences or it is just the by-product of a trivial payment choice set.

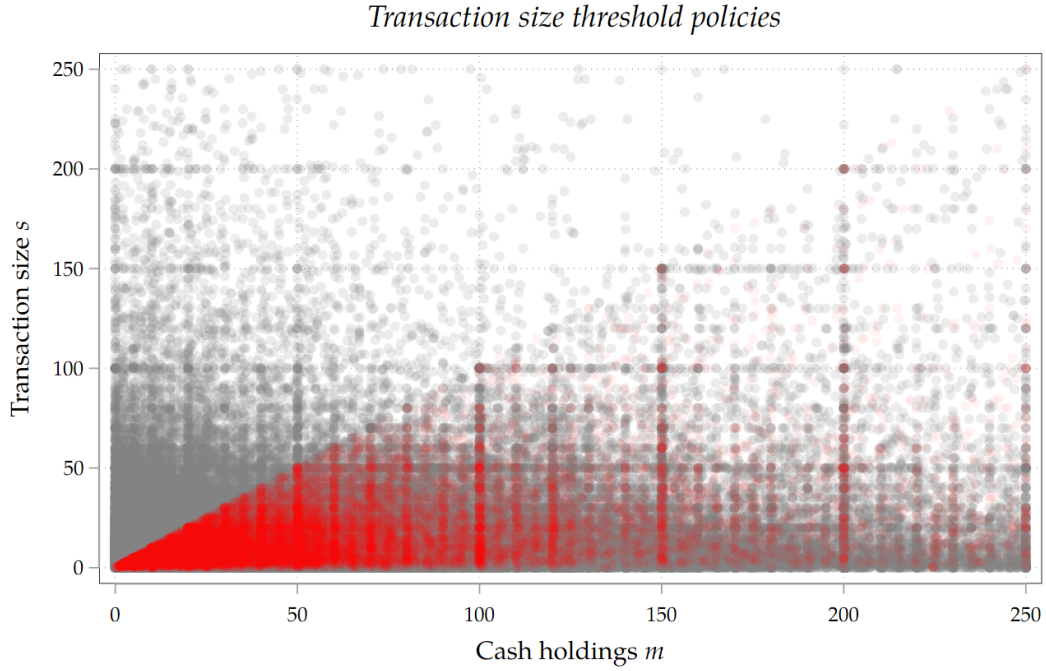
2.2 Payment choice models: a re-examination

The main goal of this Section is to re-examine the predictions of existing theoretical models of payment choice, to see if their results can rationalize payment behavior we see in the data. To this end, the ability to exactly observe payment choice sets is crucial: only voluntary payments are informative on agents' decision rules. Forced cash and cashless transactions, instead, cannot inform us on the determinants of payment choices.

Before analyzing whether existing models of payment choice are able to rationalize observed behavior, I need to briefly describe the key features of these frameworks. Most models of payment choice have a common element: they usually are enhanced versions of the standard cash management model by [Baumol \(1952\)](#) and [Tobin \(1956\)](#). In this basic model, agents must finance an exogenous, deterministic expenditure stream c using cash. They want to avoid holding too much cash on hand m , since it has a flow opportunity cost Rm ; at the same time, they want to avoid holding too little money, as withdrawing it entails a fixed cost η . This gives rise to a problem whose solution is the famous *square-root rule*, a standard formula in cash management theory that gives the optimal withdrawal: agents withdraw m^* and they wait their cash balances to reach zero; then, the cycle starts again. As payment method choices and cash management decisions are strictly related and jointly determined (I will provide evidence about this in [Subsection 2.3](#)), the most natural way to model payments is to embed a cash/cashless choice into a cash management framework similar to the one just described.

The first paper to do so is [Whitesell \(1989\)](#), where a static Baumol-Tobin model is augmented with a distribution of transaction sizes and a choice among cash and cashless payments. He finds that it is optimal to pay by cash whenever the size of the transaction is smaller than a given threshold $\bar{s}$, and to pay using cards otherwise. These decision rules (called *transaction size threshold* policies) seem to be consistent with the widely

FIGURE 1: Whitesell (1989)'s transaction size threshold policies and the data.

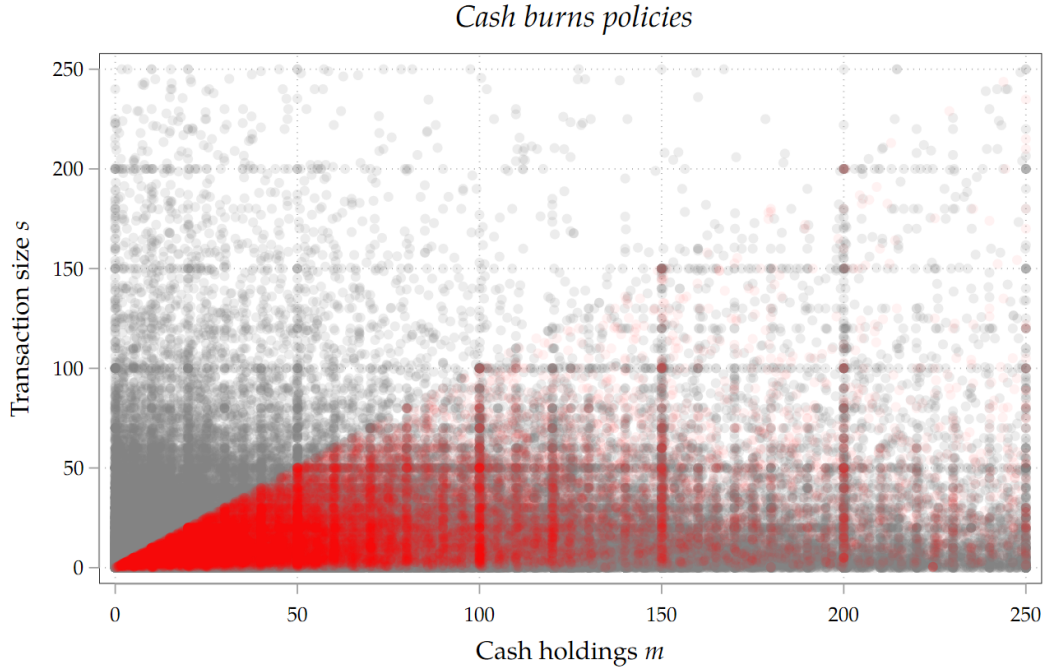


Note: Observations flagged in red are not coherent with transaction size threshold policies, even if one allows for $\bar{s}$ to be heterogeneous across agents. These are transactions where both payment options were available and either i) the agent used cash to settle a transaction bigger than her largest voluntary cashless payment, or ii) the agent used a cashless payment method to settle a transaction smaller than her smallest voluntary cash payment. Source: ECB SUCH (2016) and SPACE (2019) Data

observed phenomenon that people settle larger transactions using cashless payments than they do using cash, a behavior observed both in early studies on payment choices (Boeschoten and Fase (1989), Boeschoten (1992)) and in more recent papers that made use of high quality microdata (Klee (2008), Bagnall et al. (2016), Wakamori and Welte (2017), Arango-Arango et al. (2018)). It is possible to test the predictions of Whitesell (1989) using SUCH/SPACE data, where transaction sizes and payment methods chosen are directly observed. Since I observe more than one payment (sometimes as much as eight) for a large share of people, it is possible to test transaction size threshold policies even allowing the threshold $\bar{s}$ to be heterogeneous. If thresholds are individual-specific and given by $\bar{s}_i$, according to Whitesell (1989) one should never observe an individual performing a voluntary cash payment which is larger than her biggest voluntary cashless payment. Figure 1 shows that a large share of people explicitly violate transaction size threshold policies. Given that payment diaries in the data are only one day long, one can expect to see much more agents violating such rules as the observation time horizon is extended.

The model by Alvarez and Lippi (2017) is completely different. In their dynamic framework, payment method choices depend only on cash holdings m , but transaction

FIGURE 2: [Alvarez and Lippi \(2017\)](#)’s cash burns policies and the data.



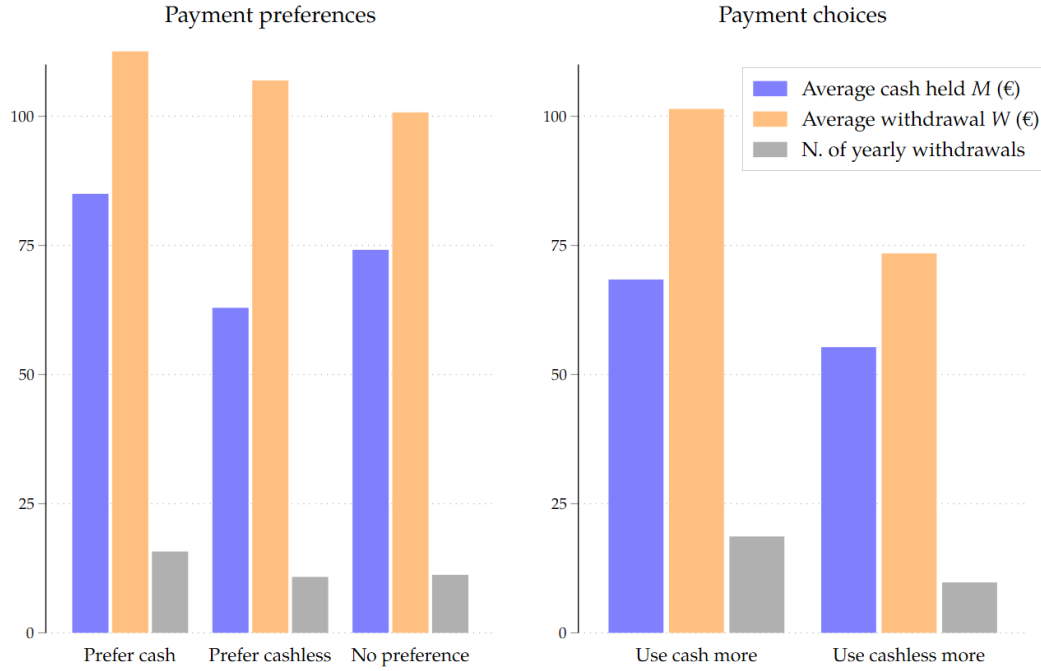
Note: Observations flagged in red are not coherent with cash burns policies. These are transactions where cashless payments were employed even if the agent had sufficiently high money holdings to settle the purchase in cash. *Source: ECB SUCH (2016) and SPACE (2019) Data*

sizes do not really play any role, as they rely on an infinitesimal and deterministic expenditure stream c , as in the standard Baumol-Tobin model. Their model’s prediction is that agents should use cash whenever they have enough. Their decision rules (called *cash burns policies*) are consistent with empirical evidence showing that the likelihood of using cash increases with the level of cash holdings ([Arango, Huynh, Fung, et al. \(2012\)](#), [Bouhdaoui and Bounie \(2012\)](#)). It is possible to test *cash burns* policies using the data: if people behaved according to such rules, one should never observe a situation where cash is enough to settle the transaction but cashless methods are employed.⁵ [Figure 2](#) shows that a large fraction of transactions cannot be rationalized by these kind of policies: often, agents prefer to pay using cashless methods even when they are not forced to do so.

There is no model based on explicit individual optimization that rationalizes voluntary cashless payments without resorting to simple threshold size policies that are contradicted by the data. Producing such a model, and studying its theoretical properties, is the aim of this work. I now exploit the information provided by SUCH and SPACE

⁵As [Alvarez and Lippi \(2017\)](#) directly point out, their model’s decision rules are only applicable for very small transaction amounts, as an infinitesimal consumption stream is assumed. Therefore, it should not be surprising that cash burn policies do not match the data when $s \gg 0$: the authors themselves acknowledge that their explanation is “complementary to that based on transaction size” (as in [Whitesell \(1989\)](#)) and “only able to address the choice of means of payment for small-sized transactions”.

FIGURE 3: Payment preferences, payment choices and cash management policies.



Note: This graph displays cash management policies across groups of individuals defined according to their payment preferences (on the left) and to their payment behavior in the day of the diary (on the right). *Source: ECB SUCH (2016) and SPACE (2019) Data.*

data in order to understand which relevant features should be included in the model.

2.3 Payment behavior: some key facts

I will now describe some features of payment behavior that can be observed using SUCH and SPACE data⁶. These stylized facts will be important to understand the basic elements that a model of payment choice should include. In particular, I argue that i) as done by previous models, jointly modeling payment and cash management decisions is crucial, and ii) a realistic model of payment choice has to feature decisions rules that depend both on cash holdings m and on the size of transactions faced s . I also provide evidence showing that imperfect cashless acceptance is a key determinant of payment choice and cash management patterns.

⁶The goal of this Section is not to give a complete overview of payment and cash management behavior in the Euro Area, but to highlight key relationships that should drive the modeling of payment choices. See respectively [Esselink and Hernández \(2017\)](#) and [ECB \(2020\)](#) for a descriptive analysis of payment and cash management patterns in 2016 and 2019.

2.3.1 The relationship with cash management

As the previous literature extensively discussed⁷, payment and cash management choices are jointly determined. There is a mutual relationship between these two decisions: on one hand, people might manage their cash in a different way based on how much they use cashless payments; on the other, when choosing on how to pay people might take into account the impact of their decision on future cash holdings.

Payment choices affect cash management decisions. Figure 3 provides further evidence of the first effect for the Euro Area in recent years. In the left panel of the Figure, I look at cash management behavior across groups defined in terms of expressed payment preferences, which agents are asked about in the survey questionnaire and which are coherent with observed choices in the diary, as will be shown later on. It is possible to see that people who reportedly prefer to pay with cash will have higher average cash holdings, withdraw more often and perform larger withdrawals than people who state they prefer to pay cashless. In the right panel, I divide people in groups based on the payment method they used more often during the day of the diary and I obtain similar results. These plots suggest that people choose their cash management strategy taking into account their payment choice habits/preferences.

Further evidence that cash management decisions are strongly influenced by payment choice policies can be found by looking at withdrawal choices. I say that a withdrawal is *needed* for an agent if her next incoming transaction is larger than her current cash balances. As long as consumers i) anticipate their next payments, and ii) like having the option to pay cash, one would expect that it is more likely that agents will withdraw when a withdrawal is needed. Moreover, one would expect this effect to be stronger for agents that prefer to pay using cash. Since I do observe withdrawal choices, I can estimate the linear probability model

$$\begin{aligned} \Pr(W_{it} = 1) = & \beta_0 + \beta_1 NeedW_{it} + \beta_2 PrefMethod_i + \\ & + \beta_3 (NeedW_{it} \times PrefMethod_i) + \\ & + \beta_4 \tilde{m}_{it} + \gamma \mathbf{X}_{it} + \eta \mathbf{D}_i + \varepsilon_{it}, \end{aligned} \quad (1)$$

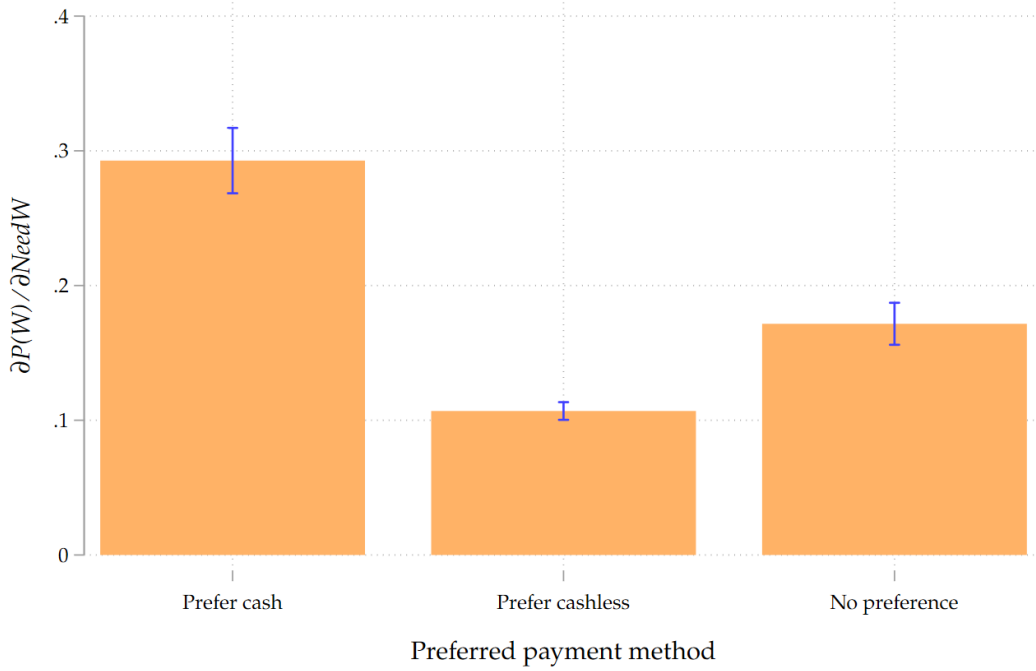
where W_{it} is an indicator variable equal to 1 if agent i withdrew money before payment t , $NeedW_{it}$ is the dummy for *needed withdrawals*⁸ and $PrefMethod_i$ is a categorical

⁷There is a large body of papers making the claim that cash management and payment choices should be studied jointly as they mutually affect each other. Applied papers making this claim are [Bagnall et al. \(2016\)](#) and [Wang and Wolman \(2016\)](#). As said in the previous Section, theoretical payment choice models usually feature a cash management problem.

⁸Formally, let s_{it} be the next transaction size and $\tilde{m}_{it}$ be current (pre-adjustment) cash holdings. Then I say that for agent i a withdrawal is *needed* before payment t if $NeedW_{it} = 1$, where

$$NeedW_{it} = \begin{cases} 0 & \text{if } \tilde{m}_{it} \geq s_{it}, \\ 1 & \text{if } \tilde{m}_{it} < s_{it}, \end{cases}$$

FIGURE 4: Effect of having insufficient cash holdings on the probability of withdrawal, by payment preferences: average marginal effects from model (1).

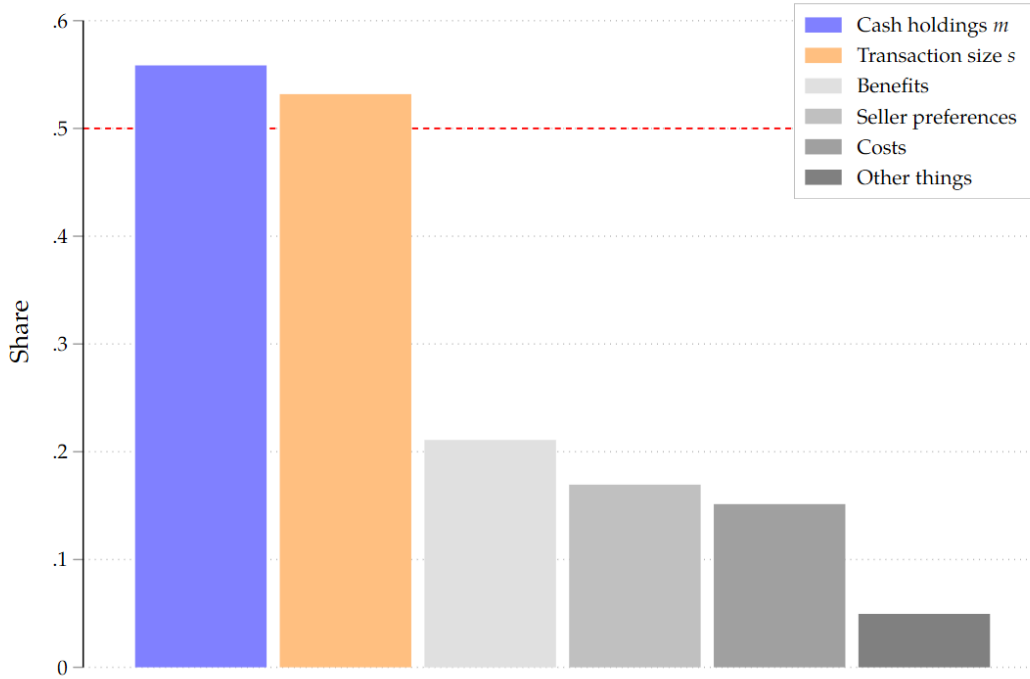


Note: This graph displays the average marginal effect of not having sufficient cash to carry out the incoming transaction ($\text{Need}W = 1$) on the probability of withdrawing money before the transaction itself ($\text{Pr}(W)$), estimated from model (1). Marginal effects are broken down by payment preferences. Blue lines represent 95% confidence intervals. Source: ECB SUCH (2016) and SPACE (2019) Data.

variable representing payment preferences. I also control for pre-adjustment money holdings $\tilde{m}_{it}$ (as agents with lower money balances are more likely to withdraw independently of cash being sufficient or not to carry out the next transaction) and for a set of transaction and individual specific covariates. In Figure 4 I plot the marginal effects of $\text{Need}W$ on W . Results show that agents that have insufficient cash withdraw more often. The effect is large: the baseline probability to withdraw money is around 2%, and it increases from 8 to 30 percentage points when cash is insufficient to carry out the incoming transaction. Moreover, there is great heterogeneity across groups defined in terms of payment preferences: the effect is almost three times stronger for agents who prefer to settle their transactions in cash with respect to those that like more paying with cashless instruments. These results again indicate that agents take into account their means of payment choices when managing their cash holdings, providing further evidence that jointly modeling the two decisions is crucial.

Cash management affects payment choices. On the other hand, people could take into account expected cash management costs when deciding on how to pay, if they have both options (cash and cashless) available. To see this, first notice that from a cash management perspective, the two payment methods are inherently different: when holding

FIGURE 5: Determinants of consumers' choice of payment instrument at the point of sale.

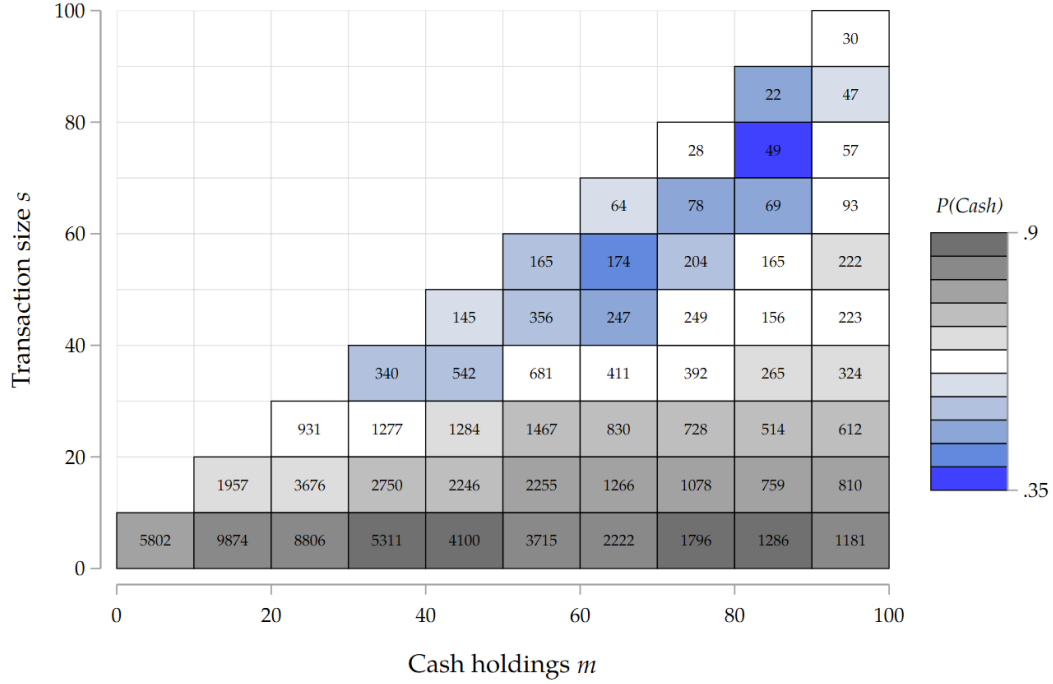


Note: This graph displays the shares of people reporting that a certain factor influences their payment decision. The question respondents answered was: “Which of the following influences your decision to pay with cash or card or other non-cash payment methods?”. Multiple responses are possible. Source: ECB SUCH (2016) Data.

m units of cash, paying a transaction of size s using a cashless instrument leaves cash balances unchanged ($m' = m$), whereas using cash depletes them ($m' = m - s$). When facing an opportunity to pay with both methods, it is natural to think that agents will weigh the static cost of performing a cashless payment (κ) with the dynamic gain associated with higher future cash balances: since the dynamic gain will depend on both m and s ($m' = m - s$ in case of a cash payment), it makes sense that these are important determinants of their choice. Fortunately, in the SUCH questionnaire people were directly asked about this, so I can test if they really care about m and s when making a payment method decision. I do so in Figure 5, where I plot the shares of agents that say if certain factors (among which m and s) affect their means of payments choice. We see that a large share of agents (more than 50%) takes into account these two factors when paying. We also see that compared to other factors, the size of the withdrawal and the size of the transaction are perceived as much more crucial for choosing on how to pay.

The fact that both m and s matter for means of payments decisions (which is consistent with payment choices being influenced by cash management concerns) is confirmed by data on actual transactions. In Figure 6 I display the share of transactions settled in

FIGURE 6: Share of cash payments for different m and s .

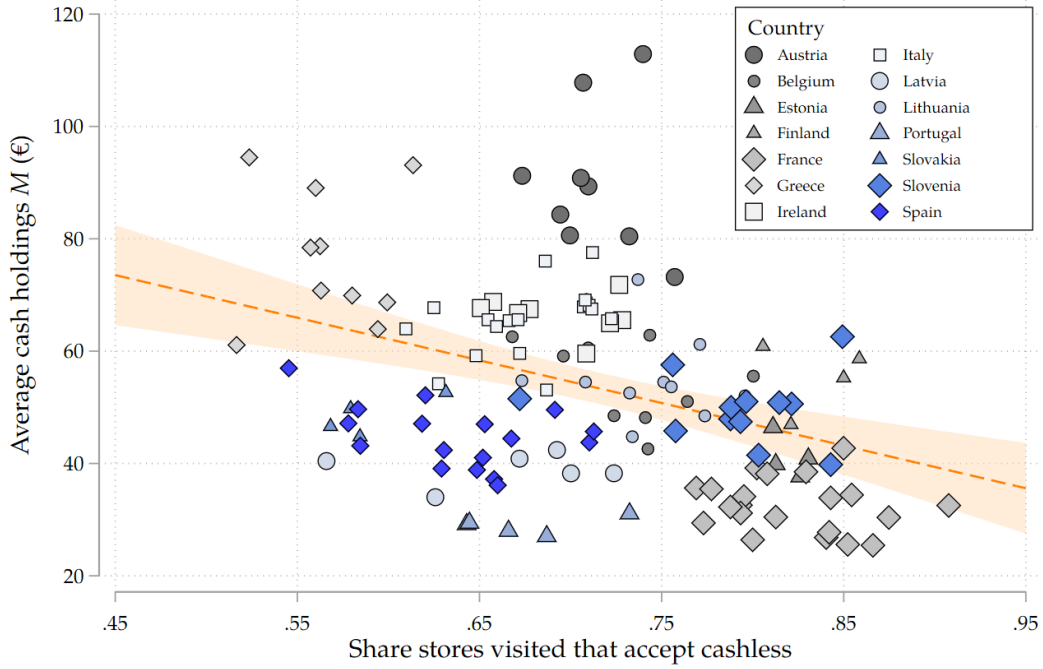


Note: This graph displays the shares of people paying with cash (when both payment options are possible) for bins defined in terms of cash holdings at payment (m) and transaction size faced (s). The share of people paying by cash is represented by the color of each cell, whereas the numbers displayed on cell denote the number of observations falling in that particular (m, s) bin. I focus on transactions where m and s are smaller or equal than 100 euros to avoid having cells with a very small number of observations. Source: ECB SUCH (2016) and SPACE (2019) Data.

cash for different values of m and s , in situations where both payment methods were accessible to consumers. Several interesting patterns emerge. First, it seems that for any given level of s , agents tend to use cash more when m is larger. Second, keeping m fixed, it seems that agents tend to use cash more for low values of s (where cash usage predominates) than for high values. Finally, when s is very close to m (on the diagonal) cashless is employed much more often. This seem to suggest that agents want to avoid using cash when this leads to a depletion of their money holdings, if they can avoid doing so. This effect is more pronounced when cash holdings are at an intermediate level (around 40-60 euros) with respect to situations where cash balances are low. These features of the data are consistent with the theory that payment choices are, at least to some extent, driven by the desire to minimize expected cash management costs: if agents desired to hold some optimal level of cash holdings m^* , they will be more inclined to pay by cash when the cash balances are far from m^* (for instance, if they are very low) than when they are close to it.

The fact that both m and s are key determinants of payment method decisions offers a possible explanation for why the models by Whitesell (1989) and Alvarez and Lippi

FIGURE 7: Average cash holdings and cashless acceptance rates in different Euro Area NUTS 2 regions.



Note: This graph plots, for each Euro Area region (NUTS 2 classification), average cash holdings in that region against the regional share of shops that accept cashless payments. A linear fit with 95% confidence intervals is overlaid to the graph. *Source: ECB SUCH (2016) Data.*

(2017) are unable to rationalize a large portion of the data: both frameworks do not feature decision rules that depend on both variables. In Whitesell (1989), as the model is static, agents' decisions are taken at the start of the period: there is way of keeping track of cash balances, and choices do not depend on cash on hand m . In a completely opposite fashion, Alvarez and Lippi (2017) do not take into account heterogeneity in transaction sizes s and their effects on payment choices. In the model I will present in Section 3, dependence of means of payment choice on m and s will be explicitly modeled in a dynamic cash management framework.

2.3.2 The role of card acceptance

Most existing models of payment choices and cash management do not take into account the fact that cashless methods are not universally accepted.⁹ This omission could lead to theoretical results which are not coherent with observed behavior, for two reasons: i) imperfect cashless acceptance introduces a precautionary motive for holding

⁹In Alvarez and Lippi (2017), the authors explicitly acknowledge the importance of cashless acceptance (and including transaction sizes too) for modeling cash/cashless choices, saying that "future models might benefit by unifying those aspects into a single model and quantify the relative importance of each of these frictions by using the relevant micro data".

cash; ii) payment choices are influenced by card acceptance levels. I now provide evidence on these mechanisms.

Acceptance rates and cash holdings. If cards are not universally accepted as a means of payment, consumers know that they may encounter a situation in which i) they don't have enough cash with them to settle a transaction; ii) the store does not accept their cashless payment method. One would expect that, in response to this, agents will optimally hold more cash in order to relax their cash in advance constraints. In other words, where cashless acceptance rates are lower, there is a stronger *precautionary motive* for holding cash. [Figure 7](#) suggests that this mechanism is actually in place. In the Figure, I plot average cash holdings and cashless acceptance rates for each NUTS 2 level region in SUCH data. The plot shows that in regions with low cashless acceptance, agents hold more cash. This seem to confirm that there exist a precautionary motive for holding cash related to uncertainty over future acceptance of alternative payment methods.

Acceptance rates and payment choices. Perceived rates of acceptance of cashless instruments may also affect payment behavior. Agents could decide on how to pay based on acceptance rates they expect to encounter in future payments. To see why, think of two consumers, A and N , both endowed with a cashless instrument (say a debit card). Both prefer to settle their purchases in cash, but still use the card when they run out of cash. A lives in an area where card acceptance is universal; N lives in an area where a large share of shops do not accept her payment card. Suppose that both consumers face a transaction in which they can use whichever payment instrument they like: given that holding cash has a higher precautionary value for N (she is not sure of being able to use the card if she runs out of cash), we expect her to use cashless payments more frequently in such circumstances in order to preserve cash for her incoming payments. I test this hypothesis by estimating the linear probability model

$$\begin{aligned} \Pr(Cash_{ijt} = 1) = & \beta_0 + \beta_1 AccRate_j + \beta_2 PrefMethod_i + \\ & + \beta_3 (AccRate_j \times PrefMethod_i) + \\ & + \beta_4 m_{it} + \beta_5 s_{it} + \beta_6 W_{it} + \gamma \mathbf{X}_{it} + \eta \mathbf{D}_i + \varepsilon_{it}, \end{aligned} \quad (2)$$

where $AccRate_j$ is the share of stores accepting cashless payments in the geographical area j , identified by the combination of a NUTS-2 level region and a urban/rural dummy. I control for key determinants of choices at POS, such as cash holdings m_{it} , transaction size s_{it} and a dummy equal to one if the agent has withdrawn cash right before the payment (W_{it}). Other demographic and transaction-specific controls are also included. Estimation results are displayed in [Table 2](#). When no controls are included and all the observations are taken into account, the relationship between cashless acceptance and the likelihood of a cash payment seems strongly negative. Part of this effect, however, is due to selection: controlling for preferences (it is more likely

TABLE 2: Estimation results for model (2).

	<i>Dependent variable: Paid cash</i>			
	(1)	(2)	(3)	(4)
Cashless accept. (%)	-0.0095*** (0.00022)	-0.0035*** (0.00074)	-0.0010 (0.0011)	0.0030* (0.0012)
Cash holdings m	0.094*** (0.0019)	0.076*** (0.0019)	0.020*** (0.0027)	0.020*** (0.0027)
Transaction size s	-0.14*** (0.0013)	-0.13*** (0.0014)	-0.093*** (0.0023)	-0.093*** (0.0023)
Withdrawn before payment	0.38*** (0.0048)	0.37*** (0.0063)	0.35*** (0.0082)	0.35*** (0.0081)
Prefer cashless		-0.24*** (0.0044)	-0.28*** (0.0058)	0.14** (0.054)
No preference		-0.10*** (0.0047)	-0.11*** (0.0061)	0.17** (0.061)
Prefer cashless $\times$ Cashless accept. (%)				-0.0056*** (0.00071)
No preference $\times$ Cashless accept. (%)				-0.0037*** (0.00081)
Observations	53987	47677	28751	28751
Robust SE	✓	✓	✓	✓
Controls		✓	✓	✓
Only voluntary payments			✓	✓
Interaction				✓

Note: In addition to controls shown, I also include demographics (sex, age, education, employment status, income, household size, internet usage), transaction-specific information (day of the week, month, type of purchase/store), region, population density (urban/rural) and year fixed effects. I only use 2019 data as 2016 data does not contain the urban/rural information, which is crucial for acceptance rates. *Source: SPACE (2019) Data.*

that people prefer cashless, and therefore use it, in regions with high card acceptance) the negative effect shrinks by two thirds in magnitude. By focusing only on voluntary payments, where both payment methods were available to the consumer (as it is much more likely that cashless was not even an option in region with low card acceptance rates), the observed association between acceptance rates and payment choices is no longer significant. Breaking down the relationship across preference groups, we see that for people who prefer cash (the baseline group), lower levels of card acceptance are associated with smaller probabilities of using cash. In particular, a 10-percent decrease in the share of stores that accept cashless methods in the area where the agent lives is associated with a decrease in cash usage (when both options are available) of around 3%.

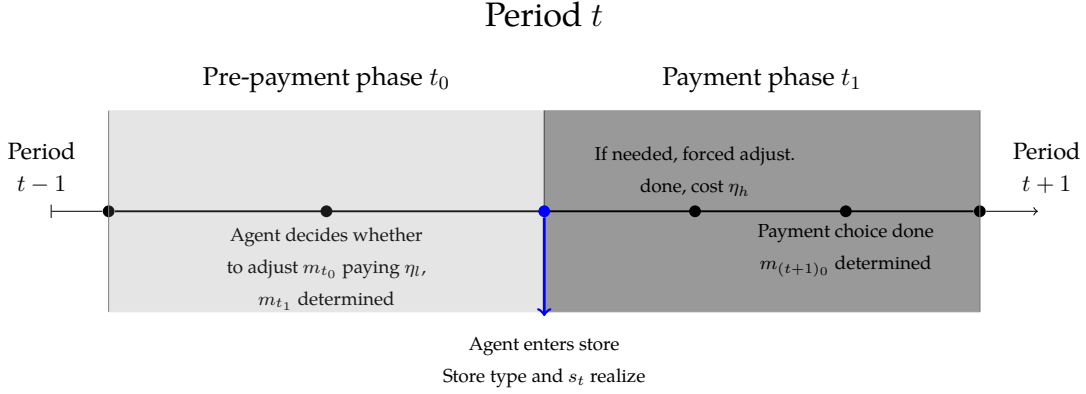
3 Theoretical Model

Why a model? A natural question to pose is why do I need a formal theoretical model of payment choice. The core idea is that, after showing features of the data which are not included in previous theories and after illustrating that many observed patterns are not explained by previous models, the goal of this paper is to illustrate theoretically that such payment patterns can be the consequence of rational choice in a dynamic cash management setting. The model will enable us to understand what are the conditions under which voluntary cashless payments are optimal, and to provide a theoretical backing for the widespread observation that smaller payments are usually settled with cash. Another benefit of laying down a formal theory is that it can be used to evaluate the effect on payment and cash management behavior of structural changes that map to model parameters, such as increases in ATM densities, improvements in cashless acceptance and so on. These are questions that cannot be answered without a structural model of behavior.

In a nutshell. Consider the problem of an agent who has access to two types of payment instruments: cash and a cashless payment method (e.g., a debit card)¹⁰. There are (potentially) three differences between cash and cashless payments: i) the cost of paying using cash or using a cashless instrument may be different; ii) the use of cashless instruments does not deplete cash balances, whereas paying by cash obviously does; iii) cash is always accepted in stores, whereas cashless instruments may not be accepted. Both withdrawing and holding cash are costly. The only way to obtain cash is to withdraw it from an ATM. The agent purchases goods and services, which he needs to satisfy exogenous needs: he can do so either through cash or using some cashless method. She wants to minimize the costs related to cash management and carrying out payments. In order to do so, she will choose the optimal amount of money to hold at each point in time, as well as the way in which she wants to settle each transaction. The underlying cash management setup is similar to the standard problem by Baumol (1952) and Tobin (1956); I add elements of payment choice models such as Whitesell (1989) (lumpy, heterogeneous and stochastic transaction sizes s) and Alvarez and Lippi (2017) (dynamic setup with choices that depend on money holdings m). Agents' payment method decisions in the model will depend on both m and s , consistently with the evidence presented in Figure 5. Another novelty of the model is the explicit inclusion of imperfect acceptance of cashless payments, which was ignored in previous frameworks.

¹⁰Throughout the paper and in the model, I never deal with the issue of card ownership. I decided to omit this margin as around 98% of people in SUCH and SPACE data have access to at least one cashless means of payment, making card ownership a second order issue for payment choices.

FIGURE 8: Timing of the model.



3.1 Setup

The timing of the model can be visualized in Figure 8. Time is discrete and indexed by t . Each period t represents a payment, and it is divided in a pre-payment phase t_0 and in a payment phase t_1 ¹¹. The agent discount the future at rate $\beta \in (0, 1)$.

In the *pre-payment phase*, agents decides whether to adjust their money holdings m_{t_0} by paying a fixed cost η_l or to keep them unchanged. In the *payment phase*, they enter a store where expenditure size s_t (which is a continuous random variable with probability density function $f(s)$ and cumulative distribution function $F(s)$) realizes. Store are of two types $j \in \{a, n\}$, with $\Pr(a) = \phi$: stores of type a accept cashless payments, while stores of type n don't. I assume that all agents own a cashless payment instrument and that they discover the size of the purchase s_t and store type j_t simultaneously, after entering the store. Let m_{t_1} be cash holdings in the payment phase of period t . In stores that accept cashless payments, the agent will be free to choose payment method whenever $s_t \leq m_{t_1}$; if $s_t > m_{t_1}$, she will clearly use cashless payments. In stores that do not accept cashless payments, the agent will pay in cash whenever $s_t \leq m_{t_1}$; when $s_t > m_{t_1}$, she will be forced to go to the closest ATM and withdraw money, paying a fixed cost η_h , potentially bigger than η_l .

I assume that cashless payments entail a fixed cost/benefit κ with respect to cash ones: the parameter κ measures taste for cash or cashless payments. As it is standard in the cash management literature, I also assume that bearing cash is costly for individuals: holding m units of cash entails a variable cost Rm . Moreover, I assume that $m \geq 0$ at each point in time: agents cannot have negative cash holdings.

¹¹The setup is similar to Arango, Bouhdaoui, et al. (2014), both concerning the division of each period in two sub-periods and the presence of discrete-sized expenditures. However, there is a major difference: in their model, the size of cash adjustments are not endogenous but randomly drawn from the observed distribution, and the authors assume the optimal payment policy is a *cash burns* policy (as in Alvarez and Lippi (2017)) with a minimum cash-holding threshold. In this model, there is explicit optimization by households.

Agent's problem. Given initial conditions m_{0_m} and s_{0_m} , the aim of the agent is to solve the discounted cost minimization problem

$$\min_{\{m_{t_0}, m_{t_1}, \tau_i, \gamma_i, \lambda_i\}_{t=0, i=0}^{\infty}} \mathbb{E}_0 \left(\sum_{t=0}^{\infty} \beta^t R m_{t_1} + \sum_{i=1}^{\infty} \beta^{\tau_i} \eta_l + \beta^{\gamma_i} \eta_h + \beta^{\lambda_i} \kappa \mid m_{0_0} \right),$$

where $\{\tau_i\}_{i=1}^{\infty}$, $\{\gamma_i\}_{i=1}^{\infty}$ and $\{\lambda_i\}_{i=1}^{\infty}$ are respectively times of morning voluntary adjustments, afternoon forced adjustments and cashless payments. The agent can minimize the cost exercising control on the amount of cash in her wallet, by deciding the timing and size of morning adjustments, the size of forced afternoon withdrawals and which payment method to use in each occasion in which she has the choice. I now illustrate the problem in simpler, dynamic programming notation.

Dynamic programming problem. Let $\hat{V}(m)$ denote the value of the problem for an agent with cash holdings m in the pre-payment phase.

$$V(m) = \min \left\{ \eta_l + \min_{\hat{m}} EV(\hat{m}), EV(m) \right\}. \quad (3)$$

The interpretation is straightforward: agents can decide whether to pay the fixed cost η_l to go to an ATM and adjust their money holdings (either by withdrawing or depositing) or to wait and keep their money balances unchanged. In the payment phase, the transaction size s realizes, and so does the acceptance policy of the shop visited by the agent. The function EV represents the expected value of the problem for an agent with money holdings m in the payment phase is given by

$$EV(m) = Rm + \phi \int_0^{+\infty} V^a(m, s) dF(s) + (1 - \phi) \int_0^{+\infty} V^n(m, s) dF(s),$$

where $\mathcal{S}$ is the support of $F(\cdot)$ and $V^a(m, s)$ ($V^n(m, s)$) respectively represent the value of facing expenditure size s when cash on hand is m in a store where cashless payments are (not) accepted. The values of being in a store that accept cashless payments and in a store that does not are respectively

$$V^a(m, s) = \begin{cases} \min \{ \kappa + \beta V(m), \beta V(m - s) \} & \text{if } s \leq m, \\ \kappa + \beta V(m) & \text{if } s > m, \end{cases} \quad (4)$$

and by

$$V^n(m, s) = \begin{cases} \beta V(m - s) & \text{if } s \leq m, \\ \eta_h + \min_{\hat{m}} \beta V(\hat{m}) & \text{if } s > m, \end{cases} \quad (5)$$

Equation 4 says that when $s \leq m$ in a store that accept cashless payments, the agent has a payment method choice. When $s > m$, instead, paying cashless is the only viable alternative. Equation 5 says that when $s \leq m$ in a store that does not accept cashless

payments, the agent will surely pay in cash; when $s > m$ the agent will have to perform a forced withdrawal at least s , but she will be allowed to choose any sum $s + \hat{m}$, with $\hat{m} \geq 0$, where $\hat{m}$ will be her starting cash balances before the next payment.

3.2 Optimal decision rules

I now examine the properties of optimal decision rules in this setup. The presence of a double discrete choice (an adjustment decision *and* a payment instrument choice) and of multiple state variables make the problem analytically intractable: even assuming a distribution of transaction sizes $F(\cdot)$, I am unable to obtain closed-form decision rules for the infinite horizon problem¹². Nonetheless, by choosing $F(\cdot)$ to match the observed distribution of payments and by solving the model numerically, it is still possible to look at policy functions to understand the fundamental properties of the model. To produce all the Figures, I solve the model assuming that $s \sim LN(\mu_s, \sigma_s)$, with μ_s and σ_s chosen to match the mean and standard deviation of the observed distribution of payments¹³.

3.2.1 Withdrawal policies

In this one-sided inventory problem, the cash management policy can be summarized by the policy function for withdrawal choices. In the pre-payment phase, the agent can either adjust her money balances to the optimal value by withdrawing cash or keep them at their value and wait. The value of adjusting is independent of m , as there are no proportional adjustment cost but only the fixed cost η_l .

Let $V^* = \min_{\hat{m}} EV(\hat{m})$. Then, by (3), the adjustment problem can be rewritten as

$$V(m) = \min \{V^* + \eta_l, EV(m)\}.$$

I now formally define the withdrawal policy function.

Definition 1. *The withdrawal policy function $(a(m), w(m))$ is given by an adjustment rule*

$$a(m) = \mathbb{1}(V^* + \eta_l < EV(m)),$$

and by a withdrawal size rule

$$w(m) = \tilde{w}(m)a(m)$$

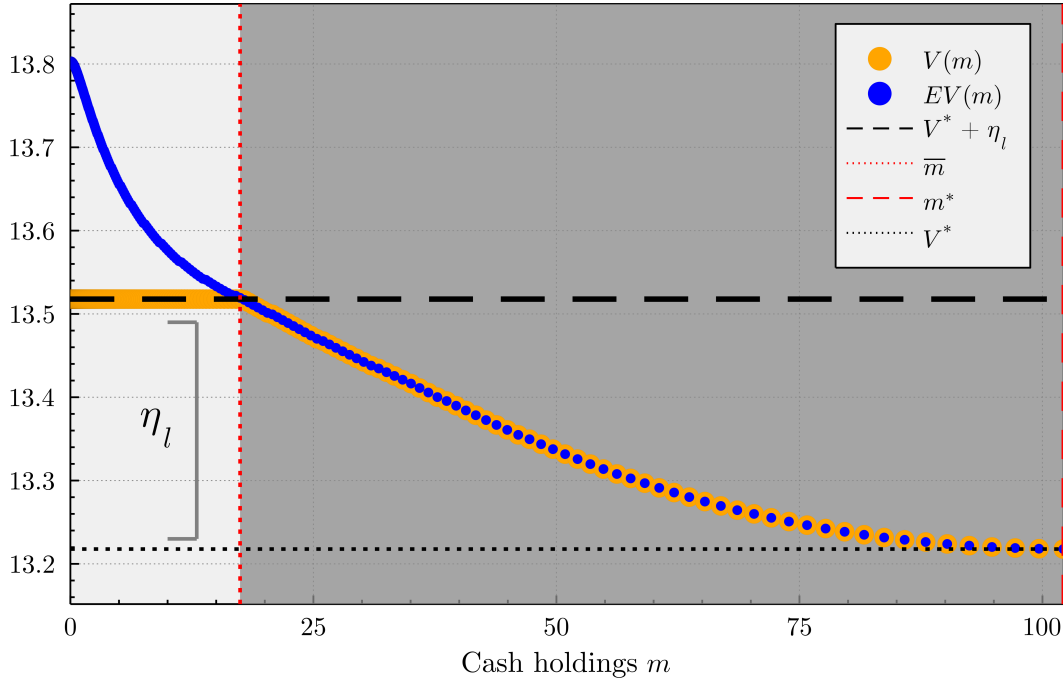
so that $a(m) = 1$ represents the choice to adjust cash balances through a withdrawal and $w(m)$ gives the withdrawal size, with $w(m) = 0$ if $a(m) = 0$.

¹²In the Appendix I obtain closed-form solutions for a simpler, two-period version of the model with full cashless acceptance.

¹³The choice of a lognormal distribution is based on Boeschoten (1992) and on SUCH data showing that a lognormal specification nicely fits the observed distribution of transaction sizes. See Appendix A.4.

FIGURE 9: Cash holdings adjustment policy function.

Parameter values: $\beta = 0.99$, $R = 0.001$, $\eta_l = 0.3$, $\eta_h = 2.0$, $\kappa = 0.1$, $\phi = 0.8$



Note: Parameters are set to their baseline values. The model is solved numerically using value function iteration on a two-dimensional logarithmic grid for (m, s) and assuming a lognormal distribution $f(s)$ for transaction sizes.

For a wide range of parameters, I find that the optimal adjustment rule is of the form that can be observed in Figure 9. The Figure shows the value of adjusting $V^* + \eta_l$, the value of non-adjusting $EV(m)$ and their lower envelope $V(m)$. For the agent, it is not optimal to adjust cash holdings for levels of money balances equal or higher than a level $\bar{m}$; when $m < \bar{m}$, agents find it optimal to withdraw money. Of course, they will withdraw enough cash to reach an optimal level m^* .

Readers familiar with the inventory-theoretical literature will immediately recognize that the adjustment policy is a classical (s, S) -rule. These decision rules are optimal in a wide class of problems with fixed adjustment costs, such as inventory management problems¹⁴: agents do not replenish inventories until their level falls below s ; when it does so, they make an order and adjust up to the level S . My cash management problem is extremely similar to a periodic-review inventory one: instead of placing orders, people withdraw cash from ATMs. For simplicity purposes, I call the renowned (s, S) -rule an $(\bar{m}, m^*)$ -rule in this application. As in standard inventory models, a necessary and sufficient condition for this kind of policy to be optimal is η_l -convexity¹⁵ of $EV(m)$.

¹⁴For many inventory models, the optimality of (s, S) policies has been proven (Scarf (1959), Iglehart (1963)). See Scarf (2005) for an overview of this class of models and for an historical perspective on the proof of optimality of (s, S) policies in inventory models.

¹⁵The following definition is due to Scarf (1959).

Proposition 1. Let $EV(m)$ be continuous and η_l -convex. Then, let

$$\begin{aligned} m^* &= \inf \arg \min_{\hat{m}} EV(\hat{m}) \\ \bar{m} &= \inf \left\{ m \in [0, m^*] \mid EV(m) \leq \eta_l + EV(m^*) \right\} \end{aligned}$$

Then the solution to (3) is a $(\bar{m}, m^*)$ -rule with $\bar{m}$ and m^* given above, and it is possible to express the value of the problem as

$$V(m) = \begin{cases} EV(m^*) + \eta_l & \text{if } m \in [0, \bar{m}), \\ EV(m) & \text{if } m \in [\bar{m}, m^*]. \end{cases}$$

The withdrawal policy function is given by the adjustment and withdrawal rules

$$\begin{aligned} a(m) &= \mathbb{1}(m < \bar{m}), \\ w(m) &= (m^* - m) \mathbb{1}(m < \bar{m}). \end{aligned}$$

Proof. The proof can be simply adapted from [Hall and Rust \(2007\)](#). However, it is a well-known result in inventory theory. ■

Since there are no exogenous inflows of cash in the model, m will never be higher than m^* . People will withdraw money and reach m^* , and they will manage their transactions optimally using cash or cashless means of payments until their cash balances reach (or go below) $\bar{m}$: then, the cycle will start again. The region $[\bar{m}, m^*]$ is sometimes defined as the *inaction region*.

In [Appendix A.3](#) I show that $EV(m)$ does not necessarily satisfy continuity and η_l -convexity for all parameter values and for all transaction size distributions: I produce two examples in which, for selected transaction size distributions, the two properties fail to hold. Since numerical solutions show that $EV(m)$ satisfies both properties when $F(\cdot)$ is a lognormal distribution (which matches the observed distribution of payments), I analyze the policy functions that arise when they are satisfied.¹⁶

Definition 2 (η_l -convexity). The function f is η_l -convex if, for all $m'' > m' > m$

$$f(m') + (m'' - m') \left[\frac{f(m') - f(m)}{m' - m} \right] \leq f(m'') + \eta_l.$$

An η_l -convex function is not necessarily convex. For instance, in [Figure 9](#) $EV(m)$ is concave for low values of m . The departure from standard convexity can be much stronger: a η_l -convex function can also have several maxima and minima. Its key feature is that, if $f^* = f(m^*)$ is the minimum of f , then: i) if $f(m) > f^* + \eta_l$ for $m > m^*$, then $f(m') \geq f(m)$ for all $m' \geq m$; ii) if $f(m) > f^* + \eta_l$ for $m < m^*$, then $f(m') \geq f(m)$ for all $m' < m$.

¹⁶ A future goal for this project is to establish conditions on the distribution of s under which η_l -convexity, and thus the optimality of $(\bar{m}, m^*)$ policies, can be proved.

3.3 Payment choice policies

The true novelty of the model is the presence of a payment choice policy. Payment choice policy functions are two-dimensional, as they depend on cash holdings m and on the size of the transaction s . Recall that agents really have a nontrivial payment choice when they are in a shop that accepts cashless payments ($j = a$) and when their cash in advance constraint does not bind ($s \leq m$). I call transactions that satisfy both conditions as *unconstrained payments*.

The value of the problem for an agent facing an unconstrained payment of size s when holding m units of cash is given by

$$V^{pay}(m, s) = \min \{ \kappa + \beta V(m), \beta V(m - s) \} = \min \{ V^{card}(m), V^{cash}(m, s) \}$$

I now formally define the policy function for the payment method decision.

Definition 3. Let the payment choice policy function be given by

$$p(m, s) = \begin{cases} 1 & \text{if } m < s, \\ \mathbb{1}(V^{card}(m) < V^{cash}(m, s)) & \text{if } m \geq s, \end{cases}$$

so that $p(m, s) = 1$ when it is optimal to pay using cashless methods.

It is not possible to give a general characterization of payment choice rules, for the same reasons highlighted in the previous Subsection. However, under some assumptions on $EV(m)$ which turn out to be satisfied for a wide range of parameters, it is possible to derive the following result, which is the core one of the paper.

Proposition 2. Let $EV(m)$ be continuous and η_l -convex. Moreover, let $EV(m)$ be non-increasing on $[\bar{m}, m^*]$. If $0 < \kappa < \beta\eta_l$, then $\exists \tilde{m} \in (\bar{m}, m^*)$, implicitly given by,

$$V(\tilde{m}) = V^* + \eta_l - \frac{\kappa}{\beta}, \quad (6)$$

such that

1. for $m \leq \tilde{m}$, for all $s \leq m$

$$p(m, s) = 0.$$

2. for any $m > \tilde{m}$ there will exist a nonempty set of transaction sizes $S(m) = (\underline{s}(m), m)$ such that

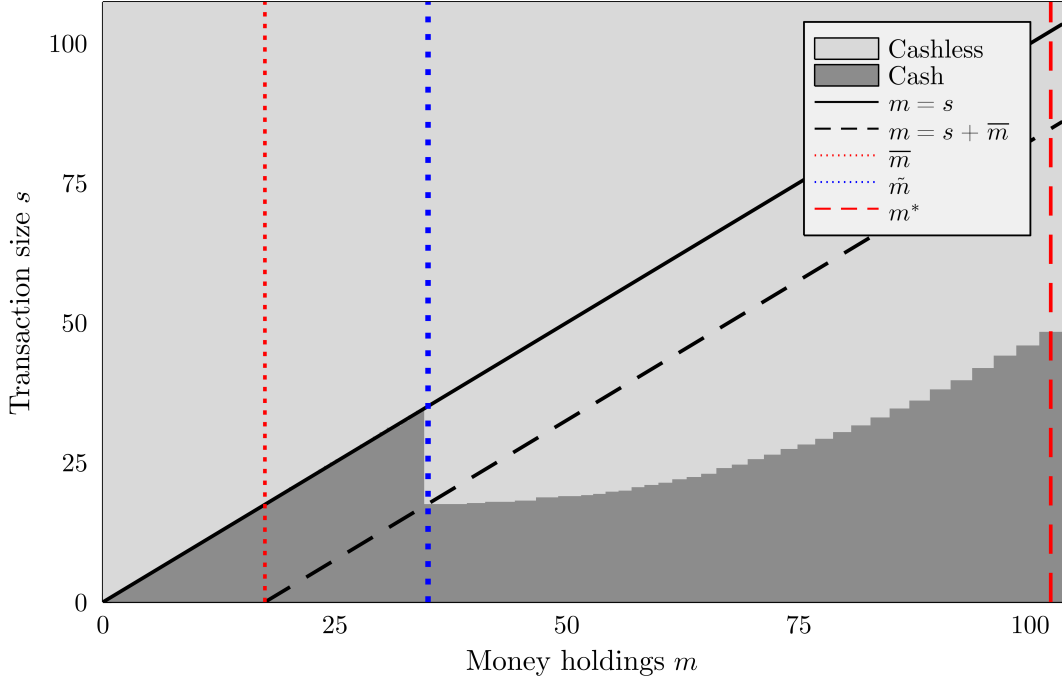
$$p(m, s) = 1, \quad \forall s \in S(m),$$

with $\underline{s}'(m) > 0$. Moreover, $\bar{s}(m) = m - \bar{m} \in S(m)$ for all $m > \tilde{m}$.

If $\kappa > \beta\eta_l$, then $p(m, s) = 0$ for all m and $s \leq m$.

FIGURE 10: Payment choice policy function.

Parameter values: $\beta = 0.99, R = 0.001, \eta_l = 0.3, \eta_h = 2.0, \kappa = 0.1, \phi = 0.8$



Note: Parameters are set to their baseline values. The model is solved numerically using value function iteration on a two-dimensional logarithmic grid for (m, s) and assuming a lognormal distribution $f(s)$ for transaction sizes. The solid black line represent the (m, s) -locus where the cash in advance constraint is binding ($m = s$): above it, there cannot be cash payments. The dashed black line represent the (m, s) -locus where paying by cash leads to the target level $\bar{m}$ ($m = s + \bar{m}$).

Proof. See [Appendix A.2](#). ■

The above Proposition describes payment behavior in the model under some assumptions on $EV(m)$ which are verified for all reasonable parametrizations when s is log-normally distributed. It says that there exist a level of cash holdings $\tilde{m}$ such that, when $m \leq \tilde{m}$, it is always optimal to pay using cash. This is because these levels of cash holdings are so far from the optimal one that it's not worth it to pay κ in order to keep them intact. In [Figure 10](#), where the policy function is plotted for baseline parameters, it's possible to see this region at the left of the dashed blue line and below the binding cash-in-advance constraint black line. When $m > \tilde{m}$, it can be optimal to pay using cashless methods in order to preserve money holdings, but if and only if the incoming transaction s is sufficiently high. This happens because the agent compares the extra cost of a cashless payments κ with the difference in expected costs $\Delta V(m, s) = \beta(V(m) - V(m - s)) < 0$: she decides to pay using cashless only if the reduction in expected future cash management cost is high enough to compensate for the extra transaction cost κ . Of course, when s is very small, $\Delta V(m, s) \simeq 0$, so it's never worth it to pay κ , as the future cash management benefits are minimal. As [Figure 10](#) shows, when s is higher than a certain threshold $\underline{s}(m)$ (which rises with m) using cash-

less is optimal, in order to keep cash balances intact: in this cases, paying using cash would decrease money holdings to a level so low that the expected discounted cost would be much higher; to avoid it, agents are willing to pay κ . From now on, I call these payment choice policies (m, s) policies.

3.3.1 Relationship with previous work

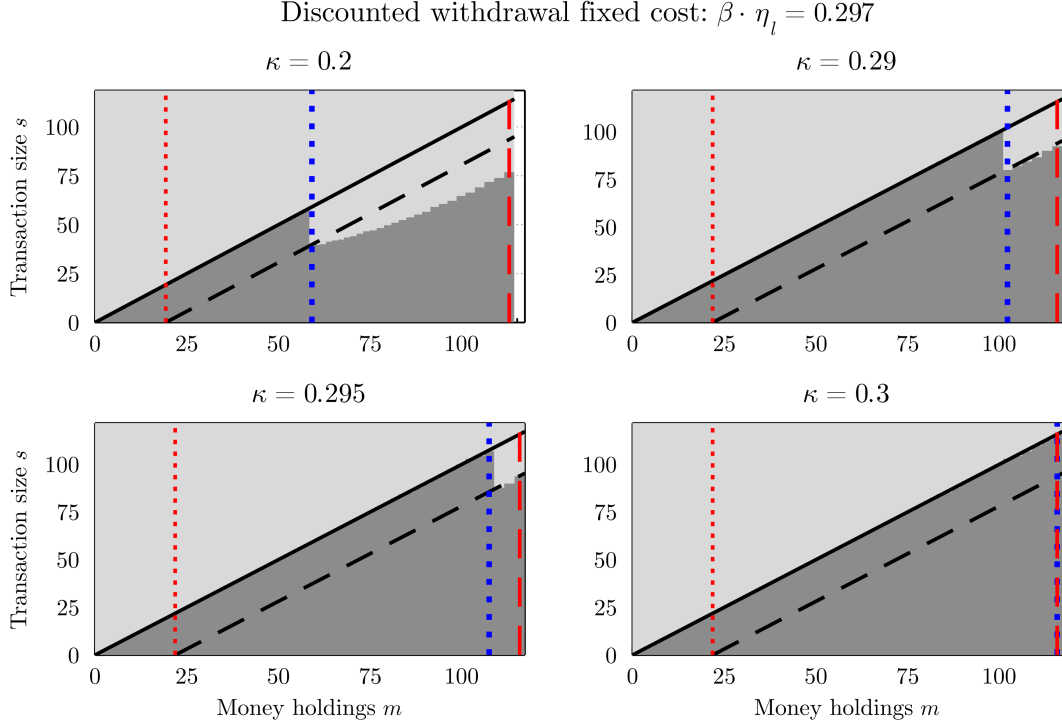
I now highlight that some of the features of decision rules found by Whitesell (1989) and Alvarez and Lippi (2017) are also present in this model, i.e., that (m, s) policies can be thought of as a more general decision rules which include features of payment choice policies discussed in previous work on the topic..

A rationalization of transaction-size thresholds. As said before, the observed negative correlation between transaction sizes and frequency of cash usage motivated Whitesell (1989) to derive transaction-size threshold policy rules for payment choices: when the size of a payment is larger than the threshold $\bar{s}$, in his model, cashless payments are optimal. However, he derived his result under two assumptions: i) agents minimize the steady-state cost of policies, with no explicit dynamic optimization or uncertainty involved.¹⁷ This creates independence of decision rules with respect to m , as decisions are taken at the start of the period; ii) the cost of using cashless methods depends on the size of the transaction, i.e., it has form $\kappa_0 + \kappa_1 s$. In the model presented in this paper, there is no need to assume that the cost of cashless payments depends on s to obtain such rules, or to arbitrarily transform an inherently dynamic stochastic problem such as sequential payment choices in a deterministic and static one. Transaction size threshold policies emerge naturally as part of (m, s) policies from the problem's dynamic structure, which makes agents internalize the consequences of their payment choices on expected cash management costs. For any level of cash holdings $m > \tilde{m}$, there exists an m -specific transaction size threshold $\underline{s}(m)$ and agents find it optimal to use cashless payments if $s > \underline{s}(m)$. This can be seen as a generalization of the theory based on the existence of a unique transaction size threshold: thresholds are not only individual specific (as they depend on policy parameters which are clearly heterogeneous, such as κ or R), they are transaction-specific, as they depend on cash holdings.

A rationalization of cash burns policies. As said above, optimal policies in Alvarez and Lippi (2017) are the so-called *cash burns* policies. People pay in cash whenever they have enough (which in their setting means any time agents have $m > 0$, as the consumption stream is infinitesimal). (m, s) policies have a similar property: when s is

¹⁷Whitesell (1989) assumes that albeit transaction sizes are not all equal, the total amount spent in each period is fixed and so is the fraction of incoming transactions of each size. Therefore, in his setup agents just choose the total amount of cash to hold, and they never run out because expenditures can never exceed their expected value: the model is deterministic. It is also static, i.e., the evolution of m in response to different sequences of transaction sizes in a given period is not modeled: given full certainty, it is in fact irrelevant, as the average cash balances in all periods (and therefore opportunity costs) will be the same independently of the sequence in which expenditures come.

FIGURE 11: Payment choice policy function as $\kappa \rightarrow \beta\eta_l$.



Note: All parameters except κ are set to their baseline values. The model is solved numerically using value function iteration on a two-dimensional logarithmic grid for (m, s) and assuming a lognormal distribution $f(s)$ for transaction sizes. The blue line represents $\tilde{m}$, which approaches m^* as $\kappa \rightarrow \beta\eta_l$.

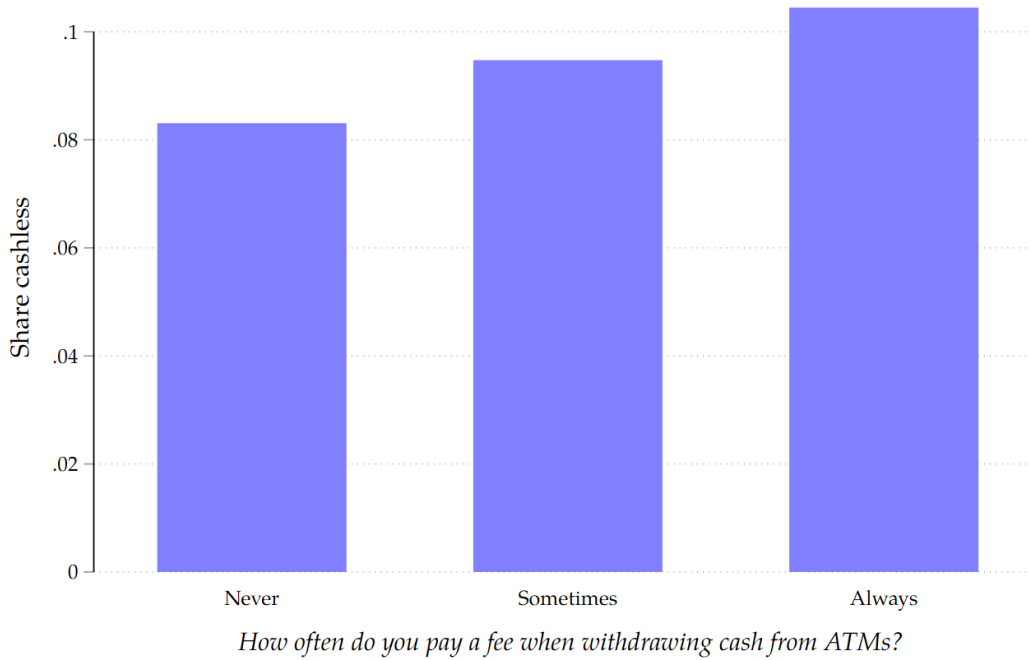
lower than some threshold (given by $\underline{s}(\tilde{m}) = \tilde{m} - \bar{m}$), it is never optimal to settle transactions using a cashless method. This is because the gain of using the card (avoiding a depletion in cash balances) is so low (as s is small, cash balances are relatively unaffected) that it's not worth it to pay the fixed cost κ . When s is extremely small, as in [Alvarez and Lippi \(2017\)](#) it is perfectly reasonable that agents prefer to use cash, as they do not push money holdings down and at the same time, they pay no cost.

4 Model Predictions and the Data

4.1 Cashless payments: a cash management tool

A novel result that emerges from optimal payment choice policies is that cashless payments can be optimal even if they are a more expensive payment method than cash ($\kappa > 0$), as long as they are a cost-effective cash management tool. This happens to be the case if and only if $\kappa < \beta\eta_l$, i.e., if it is more convenient to pay κ at the time of the payment to keep money balances intact with respect to paying with cash and then (in the next period) withdrawing. To see why it's the case, think of the opposite scenario, in which $\kappa > \beta\eta_l$: for any (m, s) , it's never convenient to pay cashless. If one wants to

FIGURE 12: Withdrawal costs and voluntary cashless payments.

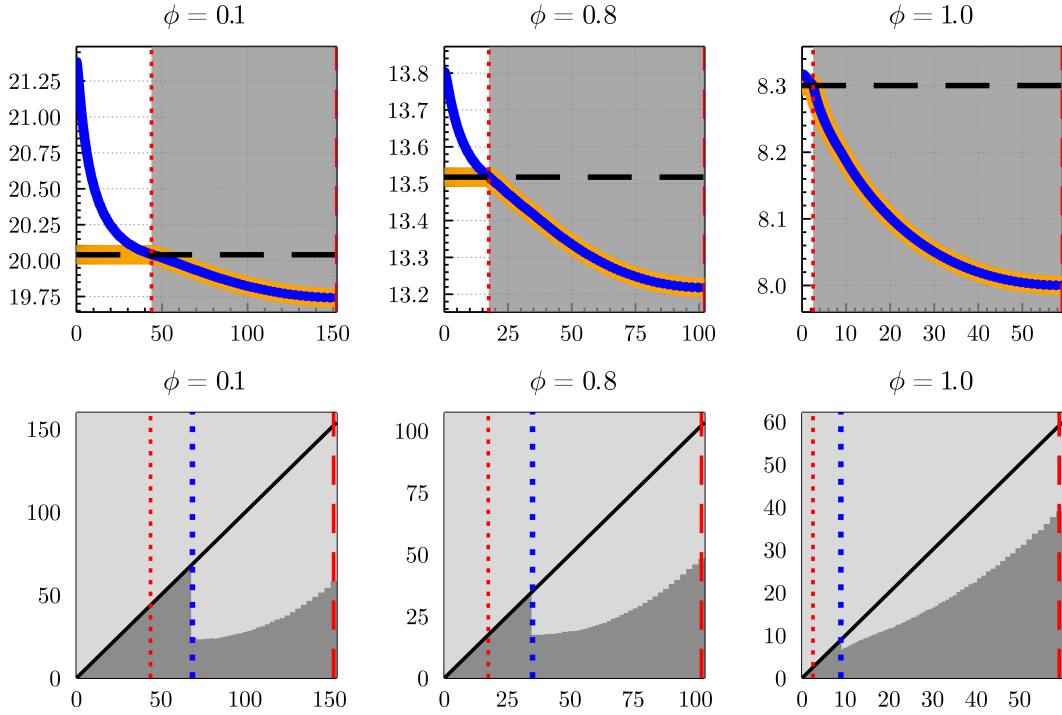


Note: Information on usual withdrawal costs faced by consumers is obtained from answers to the question “How often do you have to pay fees when withdrawing at ATMs?”. An higher likelihood of paying fees when withdrawing is a proxy of higher withdrawal costs η_l . I only consider voluntary payments, and I restrict the sample to people who reportedly prefer to use cash, to keep κ fixed while η_l varies. Source: ECB SUCH Data (2016).

keep money balances at their level, it’s cheaper to pay using cash, then withdraw to restore the previous level of cash holdings. Therefore, the model has a sharp prediction: voluntary cashless usage by agents who (ex ante) prefer cash ($\kappa > 0$) is possible only when the fixed cost of a cashless payment is lower than the discounted fixed cost of an ATM withdrawal. In Figure 11, I show this by plotting the model’s payment choice policy function for several values of κ , keeping $\beta\eta_l$ fixed.

This prediction is not easily testable as neither κ nor η_l are observed in the data. However, if one is willing to make some assumption about the relationship of κ and η_l with observables, it is possible to obtain some insights. As our measure of κ we can exploit the answer to the question about preferred payment methods. Let’s focus on agents that say they prefer cash, so that $\kappa > 0$ (as in the model) seems a reasonable assumption. Ideally, one would like to keep κ fixed and see how the share of voluntary cashless payments changes with η_l . As a proxy for η_l , we can exploit the answer to the question “How often do you have to pay fees when withdrawing at ATMs?”, which was asked in the SUCH survey (2016). Agents report if they never pay fees when withdrawing, if they sometimes have to pay or if ATM usage always entails the payment of a fee.

FIGURE 13: Policy functions for different acceptance levels ϕ .



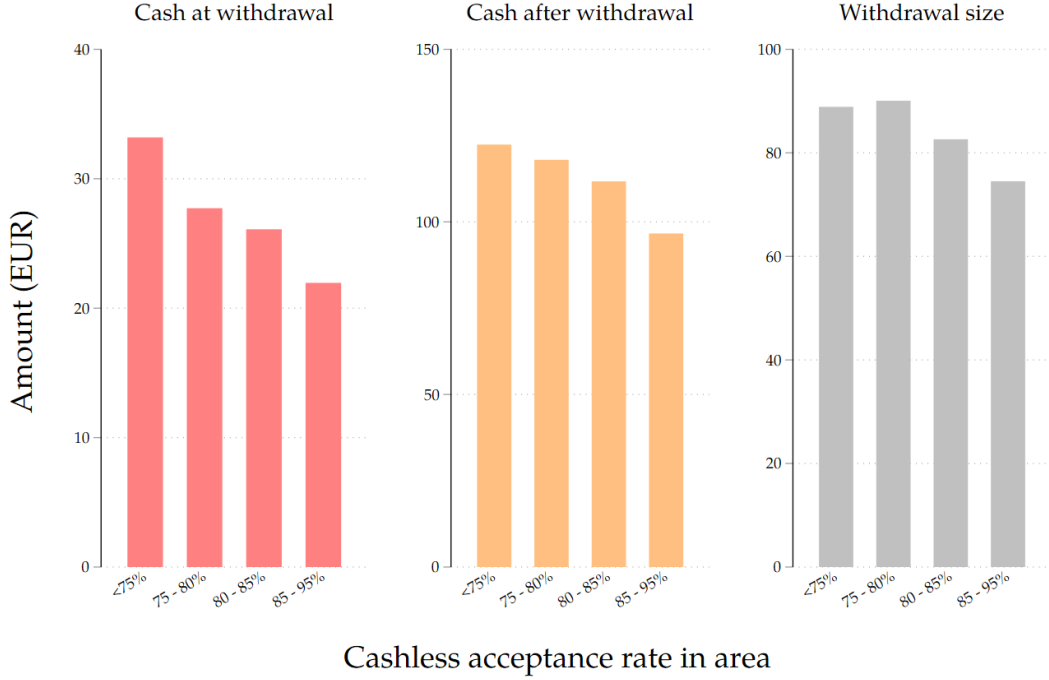
Note: All parameters except ϕ are set to their baseline values. The model is solved numerically using value function iteration on a two-dimensional logarithmic grid for (m, s) and assuming a lognormal distribution $f(s)$ for transaction sizes.

4.2 The effect of card acceptance

In the policy functions displayed before, I always kept cashless acceptance constant at the level $\phi = 0.8$. I now investigate how is cash management and payment behavior influenced by changes in expected acceptance and see if model predictions regarding these effects are consistent with empirical evidence. In Figure 12 I display the share of voluntary payments which is performed using cashless methods as a function of the proxy for η_l . Results show that higher withdrawal costs are associated with a higher share of voluntary payments being settled using cashless methods, which is consistent with the theoretical implications of the model regarding the cash management role of card payments. When agents find it harder to withdraw money, they increase card usage to preserve higher cash balances, instead of using them and replenish at ATMs more often. In Figure 13 I display withdrawal and payment choice policy functions for three levels of cashless acceptance ϕ : its baseline value $\phi = 0.8$ (which is close to the Euro Area average for 2019), very low cashless acceptance ($\phi = 0.1$) and full cashless acceptance ($\phi = 1$). Comparative statics via numerical solutions show the following effects of card acceptance levels on payment choice and cash management patterns.

1. The highest possible level of cash holdings at the time of a withdrawal $\bar{m}$ is de-

FIGURE 14: Acceptance rate and adjustment policies in the data.



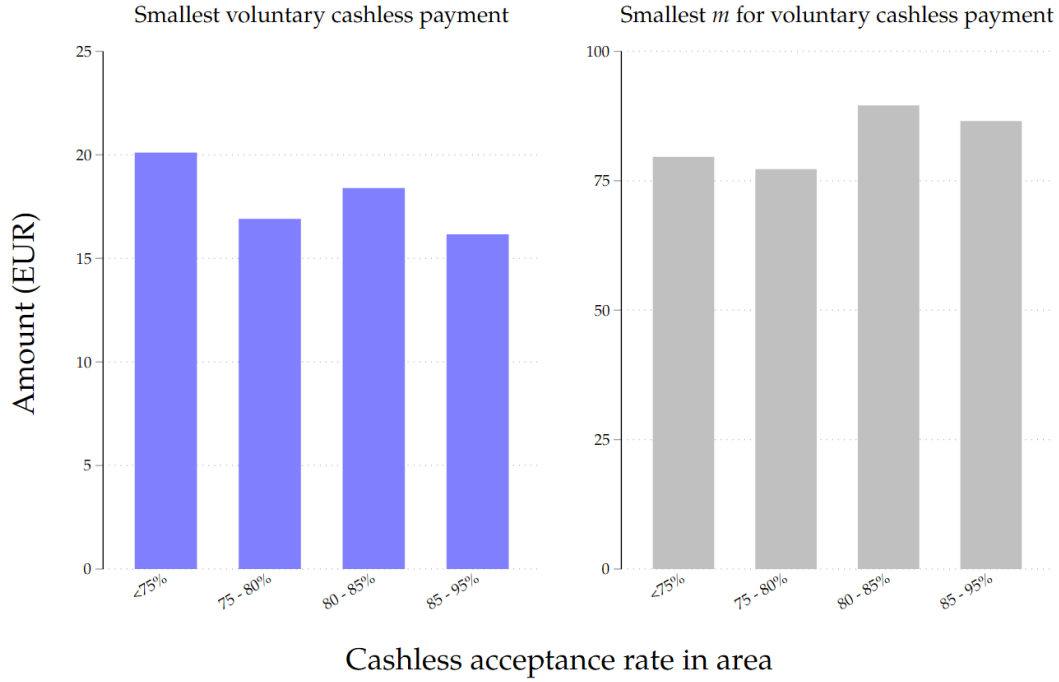
Note: All measures are based on agents' reported withdrawals. Cashless acceptance rates are computed at the region-year level. *Source: ECB SUCH (2016) and SPACE (2019) Data*

creasing in ϕ ;

2. The level of cash holdings after a withdrawal m^* is decreasing in ϕ ;
3. The smallest possible withdrawal $m^* - \bar{m}$ is decreasing in ϕ ;
4. The smallest level of cash holdings for which a voluntary cashless payment is possible $\tilde{m}$ is decreasing in ϕ .
5. The smallest voluntary cashless payment $\underline{s}(\tilde{m})$ is decreasing in ϕ .

The first three findings are consistent with imperfect acceptance inducing an extra precautionary motives for holding cash: when cards are always accepted, agents do not fear anymore to run out of cash and being forced to withdraw in case they visit a cash-only store when their money holdings are insufficient. When card acceptance is very low, they hold a significant buffer stock of cash to insure themselves against such circumstances; therefore, the minimum cash holdings they are willing to keep $\bar{m}$ will be higher, and so will be the target value of cash holdings they obtain upon withdrawing m^* . The difference $m^* - \bar{m}$ (smallest possible withdrawal) is also decreasing in ϕ : the model implies that people living in areas with higher card acceptance should make smaller withdrawals. In Figure 14 I investigate if these predictions actually hold in the data, by looking at money holdings at the time of a withdrawal (a proxy for

FIGURE 15: Acceptance rate and adjustment policies in the data.



Note: In the left panel, the outcome is the transaction size of the smallest voluntary transaction settled using cashless methods (a proxy for $\underline{s}(\tilde{m})$). In the right panel, the outcome is the smallest level of cash holdings for which a voluntary cashless payment is made (a proxy for $\tilde{m}$). Cashless acceptance rates are computed at the region-year level. *Source: ECB SUCH (2016) and SPACE (2019) Data*

$\bar{m}$), at money holdings immediately after withdrawals (a proxy for m^*) and at withdrawal sizes (a proxy for the minimum withdrawal size $m^* - \bar{m}$). The observed correlation between cash management policies and cashless acceptance rate is consistent with comparative statics from the model.

Unfortunately, there are not such good proxies for $\tilde{m}$ and $\underline{s}(\tilde{m})$ in the data. One could look at the smallest cash holdings $\hat{m}_i$ for which a voluntary cashless payment is made, and at the smallest transaction size settled voluntarily using cashless methods $\hat{s}_i$. This is what I do in Figure 15. The left panel shows that the minimum transaction size for which agents pay cashless is decreasing in the level of acceptance, as predicted by the model. The result for the smallest level of cash holdings are instead not in line with the model, showing a small positive association. Further investigation on this is needed, maybe making use of regression analysis instead of simple descriptive statistics to see if factors correlated with high card acceptance (such as higher income, expenditure levels and so on) are biasing these results.

4.3 The data and (m, s) payment choice policies

As said before, the model features cash-holding specific thresholds $S_i(m)$, that could be different for any individual. If one assumes that the policy functions are individual-specific functions $p_i(m, s)$ but that they are constant across time (a reasonable assumption given that diaries cover just one day), it is possible to test some restrictions imposed by the theory. First note that, for the same m , one should never observe $p(m, s) = 1$ and $p(m, s') = 0$ for $s' > s$. We actually observe a lot of individuals facing different transaction sizes with the same m (this is quite natural, as when they pay cashless m will be the same for the next payment in the absence of withdrawals/deposits). However, the shape of payment policy functions creates a more general, testable set of restrictions, based on the following Proposition.

Proposition 3. *If the assumptions of Proposition 2 are satisfied, then for any $m > \tilde{m}$:*

1. *if $p(m, s) = 0$, then $p(m', s') = 0$ for any $m' \geq m, s' \leq s$.*
2. *if $p(m, s) = 1$, then $p(m', s') = 1$ for any $m' \in [\max\{\tilde{m}, s\}, m], s' \geq s$.*

Proof. Follows from the fact that $\underline{s}(m)$ is increasing in m . ■

Notice that Proposition 3 nests the statement that for the same m , one should never observe a cash payment bigger than a cashless payment, for the same individual. The implications of Proposition 3, however, are hard to test for one reason: we do not observe $\tilde{m}_i$. However, given observed voluntary payments¹⁸ $\mathbf{P} = \{P_{it}\}_{i=1, \dots, N}^{t=1, \dots, T_i} = \{m_{it}, s_{it}, p_{it}, v_{it}\}_{i=1, \dots, N}^{t=1, \dots, T_i}$ I can put bounds on $\tilde{m}_i$ and determine if choices are coherent with the model's implications. I do it exploiting the following recipe.

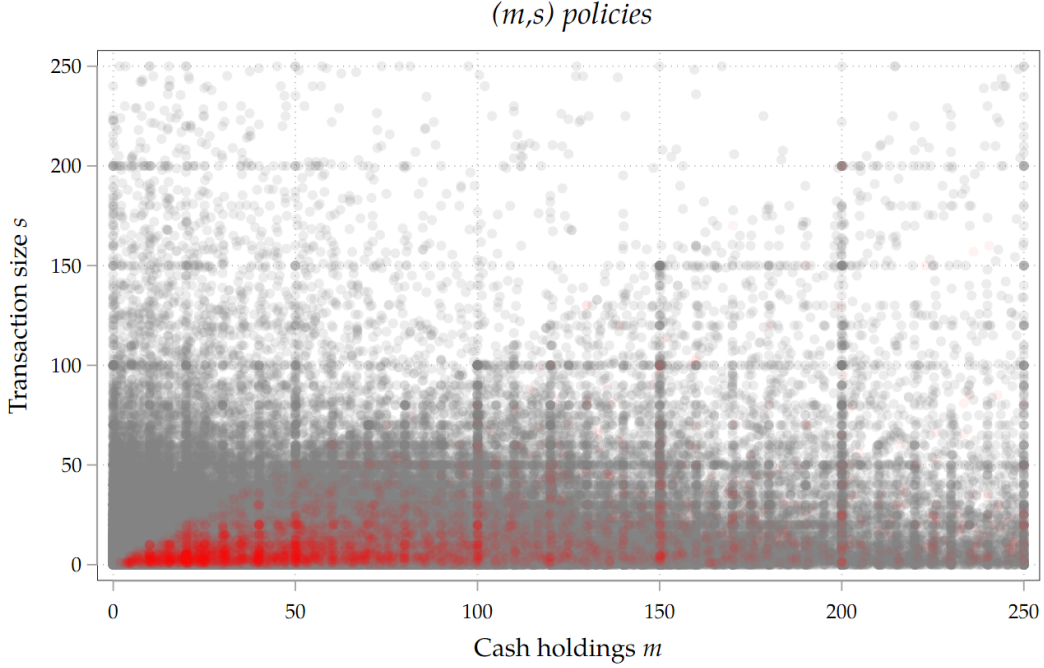
Recipe 1. *For each individual i , let $\hat{m}_i$ be the smallest level of cash holdings m_{it} for which a voluntary transaction is settled by using cashless methods. Clearly, it has to be the case that $\tilde{m}_i < \hat{m}_i$. We say that payment t by individual i , P_{it} , is inconsistent with the model if it is voluntary ($v_{it} = 1$) and one of the following two conditions holds.*

1. *If $p_{it} = 1$ and there exist another payment $P_{ir} = (m_{ir}, s_{ir}, p_{ir}, v_{ir})$ with $v_{ir} = 1$, $m_{ir} \geq \hat{m}_i$ and $p_{ir} = 0$, such that $m_{ir} \leq m_{it}$ and $s_{ir} \geq s_{it}$.*
2. *If $p_{it} = 0$ and there exist another payment $P_{ir} = (m_{ir}, s_{ir}, p_{ir}, v_{ir})$ with $v_{ir} = 1$, $m_{ir} \geq \hat{m}_i$ and $p_{ir} = 1$, such that $\max\{\tilde{m}_i, s_{ir}\} \leq m_{it} \leq m_{ir}$ and $s_{ir} \leq s_{it}$.*

We say that payment behavior of agent i is inconsistent with the model if at least one payment P_{it} by agent i is inconsistent.

¹⁸Individuals are denoted by i , transactions by t . For each individual, I observe T_i transactions (up to eight). I already defined m_{it} and s_{it} ; p_{it} represents the payment choice ($p_{it} = 1$ is a cashless payment) and v_{it} is a dummy equal to one if the payment is voluntary.

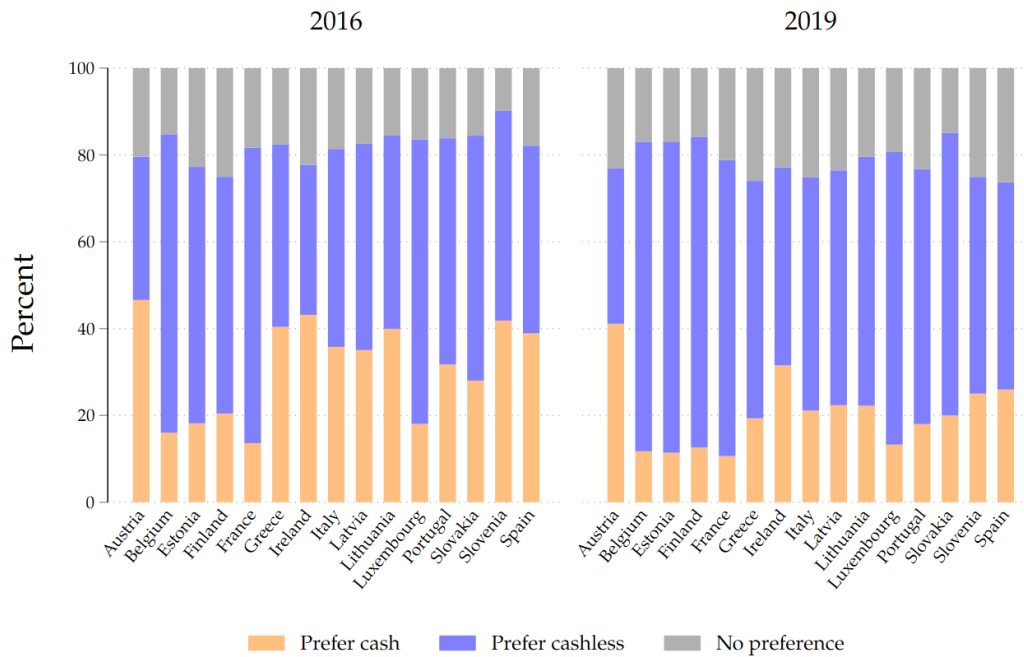
FIGURE 16: The model's (m, s) policies and the data.



Note: Observations flagged in red are all the transactions inconsistent with (m, s) policies. These payments do not satisfy the conditions of Proposition 3. Source: ECB SUCH (2016) and SPACE (2019) Data.

Both conditions are based on Proposition 3. Observations labeled as *inconsistent* violate the Proposition. Of course, a necessary condition to say if an individual has behavior which is consistent with the model is to observe at least two voluntary transactions, one of which must have been settled using cashless methods. In Figure 16 I flag in red all transactions which are inconsistent with the model given the criteria established above. Compared to Figure 1 and Figure 2, it is possible to see that a much higher share of payments are rationalized by the model, which imposes weaker restrictions than Whitesell (1989) or Alvarez and Lippi (2017). Despite that, it is still clear that the model is not able to rationalize all payments, especially at low transaction values: this is probably due to the fact that, in the model, the relative cost of each payment method is fixed (given by κ) whereas in reality it may vary a lot across transactions for reasons which cannot be incorporated easily into the model. For instance, as shown in Figure 5, agents often take into account seller preferences when deciding on how to pay. If seller preferences are correlated with transaction values and they influence the agent's utility from choosing a certain payment method, then κ is not fixed anymore but depends on s . In general, any quantitative model of payment choice built on the framework presented here should allow for payment-method specific utility shocks, to model unobserved determinants of decisions in the style of Rust (1987).

FIGURE 17: Payment preferences.



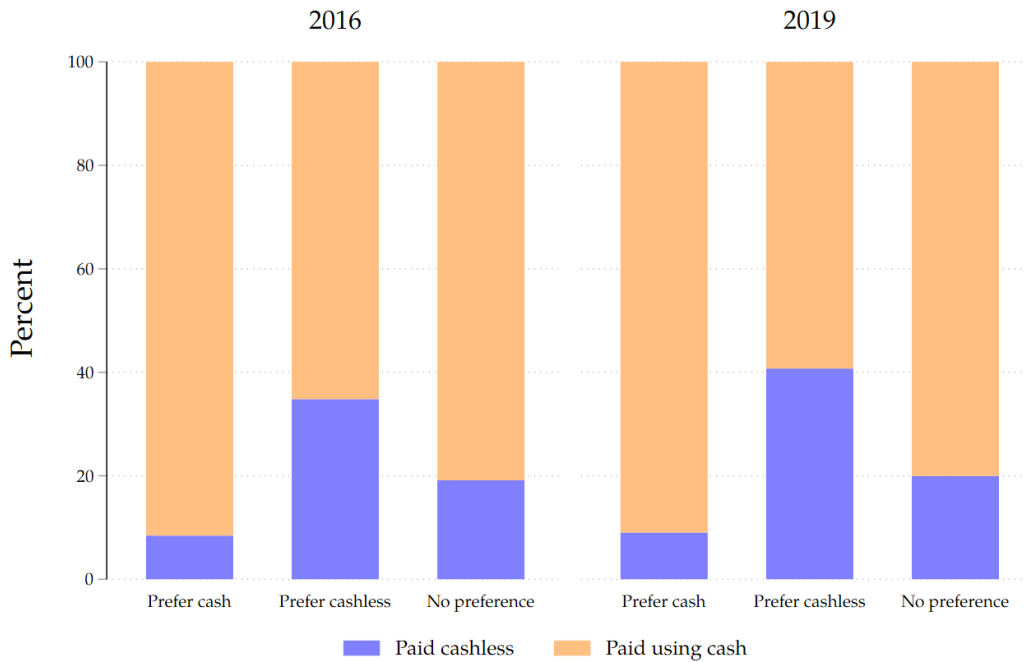
Note: This graph plots the answers to the questionnaire question “If you were offered various payment methods in a shop, what would be your preference?” for all countries in the survey, both for SUCH (2016) and for SPACE (2019). The options were cash, a generic cashless option and the absence of a preferred payment method. The plot is based on data for a total of 65,858 respondents. Source: ECB SUCH (2016) and SPACE (2019) Data.

4.4 Rationalizing a payment choice puzzle

Both in the SUCH and in the SPACE questionnaires, respondents were asked about their preferred payment method between cash and cashless methods. They could pick one of the following responses: *cash*, *cashless* and *no preference*. In what follows, I sometimes refer to people in the first category as *cash lovers* and to people in the second as *cashless lovers*. Answers to this question, for both surveys/years, are reported in Figure 17. As the Figure shows, the appreciation for cashless payments has been growing in recent years and they are now preferred to cash in many countries¹⁹. One can focus on the relationship between these responses and actual choices at the point of sale. Being able to observe the payment choice set of agents at each point in time, I concentrate on situations in which consumers had both payment options available. In those situations, one could think that agents would behave according to the preference ordering stated in the questionnaire, i.e., that they would pay with cash if they reportedly prefer doing so or with a cashless instrument in the opposite case. On the other hand, we have no prior on the behavior of agents that reported no absolute preference for one payment

¹⁹In 2019 data, Austria is the only country in which there were more respondents saying that they prefer cash than respondents saying that they prefer cashless payments.

FIGURE 18: Stated preferences and actual choices at POS when both payment options are available: the *payment choice puzzle*.



Note: This graph summarizes payment choices of agents in situations when both payment options were available (cash and cashless methods were both accepted, the agent had enough cash in her wallet and she had access to a cashless payment instrument). Choices are broken down as a function of stated preferences (prefers cash, prefers cashless, no preference). This panel is based on a total of 56,395 transactions. Source: ECB SUCH (2016) and SPACE (2019) Data.

option over the other.

I compare stated preferences and actual unconstrained choices in [Figure 18](#), where I restrict the sample to transactions where both payment options were available to the agent. Importantly, the Figure shows that questionnaire answers are not *cheap talk*: there is a clear correlation between stated preferences and actual choices. In situations in which they are free to choose among the two payment methods, those who said they prefer cash use it more often than people who said they like cashless payments more, with people expressing no preference placing in between. Despite being informative, [Figure 18](#) makes it clear that stated preferences are far from being a perfect predictor of behavior, i.e., that preferences expressed in a questionnaire and choices do not always match even when there are no constraints. This holds both for cash lovers and for cashless lovers.

The fact that cashless lovers settle the majority of their voluntary transactions (around 60%) in cash is not novel in the payments literature, and several explanations have been put forward. For instance, [Crujisen, Hernandez, and Jonker \(2017\)](#) exploit very detailed Dutch data to claim that payment habits are driving the persistence of cash

usage among people who state they prefer cards. A large literature has examined the impact of supply-side factors such as sellers' preferences for receiving cash payments (Arango, Huynh, and Sabetti (2015), Bagnall et al. (2016), Wakamori and Welte (2017)). Discussing the reasons behind intensive cash usage by people who report a cashless preference is not within the scope of this paper.

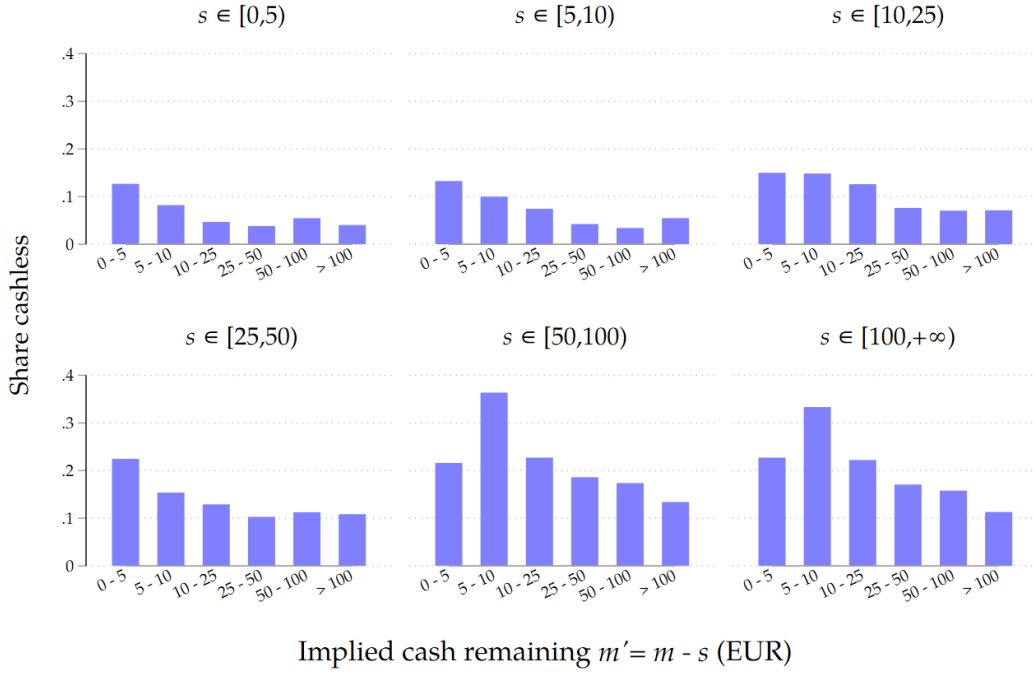
What I do focus on, instead, is a fact which has not been analyzed at all in the literature: cash lovers settle a sizeable portion (around 10%) of their voluntary transactions using cards. Here, supply-side factors and habits should not play any role, as they both go in the opposite direction (i.e., they should reinforce cash usage intensity by cash lovers). Why would agents that prefer to use cash as a payment method willingly choose to pay using cards sometimes? I call this a *payment choice puzzle*.

The data provide a possible interpretation of this puzzling fact which is consistent with the mechanisms of the theoretical model illustrated above. In the model, we showed results on payment choice policies for the case in which $\kappa > 0$: in this case, agents *are* cash lovers, as they have an ex-ante preference for cash. However, as I have showed theoretically, even agents that face a fixed cost when using the card may find it optimal to use it for their cash management needs, especially when i) they have money holdings close to their optimal level m^* , and ii) they face a transaction s that would significantly deplete their cash holdings, i.e., $m' = m - s \ll m^*$.

It is possible to test using SUCH and SPACE data if this is really the case. I can investigate whether cash lovers are more likely to perform cashless payment when implied cash holdings in the case of a cash payment ($m' = m - s$) are low than when they are at an intermediate level. This would be consistent with the model, as agents want to avoid holding too little cash and they are willing to pay cashless (even though they dislike it with respect to paying cash) to keep their money balances high.

In Figure 19 I show that the data seem to confirm this hypothesis. The Figure shows that cash lovers are more willing to settle their transaction using a card if implied cash holdings are very low, while the share of voluntary cashless transaction they make is much lower when implied cash holdings are at an intermediate or high level. This holds for any level of transaction sizes s , despite the intensity of cashless usage is way higher when s increases. It also makes sense that the effect of a decrease in m' on payment choices is more pronounced for high s : when s is small, the gain in future expected cash management costs associated with card usage is lower, as m and m' are very close; when s is large, m and m' are significantly different and therefore there is more scope for card usage, i.e., it is more likely that the benefits of using the card (lower expected cash management costs in the future) exceed its direct cost κ .

FIGURE 19: Voluntary cashless usage by cash lovers and implied cash remaining.



Note: This graph is based on transactions performed by cash lovers that had both payment options available. It documents the frequency of voluntary cashless usage by cash lovers for different values of $m' = m - s$, the level of cash remaining if the transaction is settled using cash, and for different levels of s (the size of the transaction). Source: ECB SUCH (2016) and SPACE (2019) Data.

5 Cross-Country Differences: A Calibrated Model

I now turn to assessing the model's potential to replicate relevant features of the data. The purpose of this final part of the paper is not to formally establish conditions under which identification of the model can be achieved, nor to present estimates of model parameters: for that, a quantitative model would be needed. My scope here is to show that the model can replicate some key data moments concerning payment choice and cash management, and that variation in parameters can capture the cross-country heterogeneity in observed patterns.

5.1 Payment and cash management behavior in the Euro Area

Before calibrating the model, I will now present data showing massive heterogeneity in payment and cash management behavior in the Euro Area. This will enable me to select some countries which are very different in terms of observed moments, and in the next Subsection I will investigate if the model can replicate these differences, at least qualitatively. In Table 3 I display six relevant moments on cash management and payment choices for all countries in the data in 2019, the year in which the last survey

TABLE 3: Payment choices and cash management, cross-country evidence.

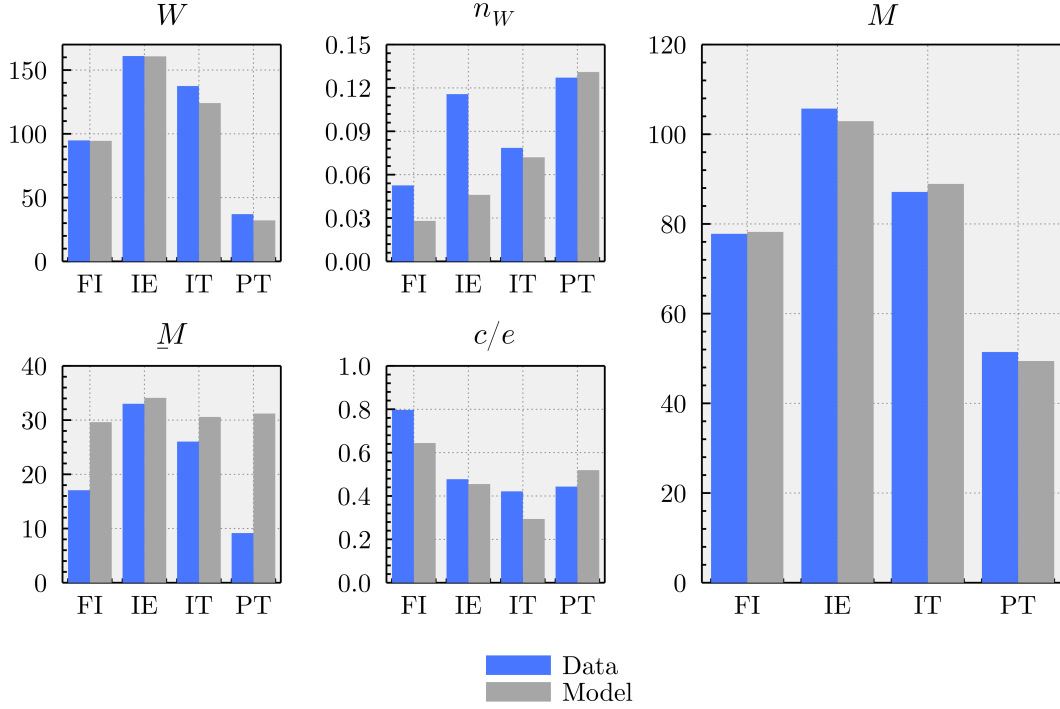
	M	W	n_W	$\underline{M}$	ϕ	c/e
Austria	122.10	101.70	0.11	18.92	0.74	0.44
Belgium	75.70	98.10	0.07	23.51	0.79	0.65
Estonia	69.33	75.38	0.05	15.38	0.86	0.73
Finland	77.79	94.86	0.05	17.06	0.92	0.80
France	54.47	65.46	0.05	18.42	0.86	0.66
Greece	110.04	123.84	0.09	36.69	0.85	0.42
Ireland	105.71	160.99	0.12	33.01	0.80	0.48
Italy	87.13	137.47	0.08	26.03	0.75	0.42
Latvia	65.15	80.08	0.07	18.07	0.76	0.57
Lithuania	87.38	74.01	0.07	14.31	0.76	0.54
Luxembourg	129.08	121.03	0.09	21.21	0.85	0.66
Portugal	51.44	37.04	0.13	9.16	0.72	0.44
Slovakia	81.07	208.01	0.08	41.86	0.68	0.48
Slovenia	80.74	95.77	0.09	28.42	0.80	0.47
Spain	95.14	98.52	0.08	32.26	0.77	0.41
Total	86.15	104.82	0.08	23.62	0.79	0.54

Note: The first four moments are related to cash management. M are average cash holdings (EUR), W is the average withdrawal, n_W is the share of people withdrawing in a given day, $\underline{M}$ is the average cash balance at the time of a withdrawal. The last two moments concern payment choice: ϕ is the level of card acceptance in the country, and c/e is the cashless share of total expenditure. *Source: SPACE (2019) Data.*

(SPACE) was carried out. In particular, regarding cash management behavior, I display average cash holdings M , average withdrawals W , the share of agents withdrawing in a given day (a proxy for withdrawal frequency: the higher it is, more often agents withdraw) and average cash at withdrawal $\underline{M}$. As for payment choices, I focus on card acceptance levels ϕ and the cashless share of expenditure c/e .

As the Table shows, there is huge heterogeneity across the Euro Area in all these dimensions. Average cash balances held by consumers range from around 50 euros (as in Portugal and France) to more than 120 euros (as in Austria or Luxembourg). In some countries, agents make huge withdrawals (compared to their average holdings), as in Slovakia; people from Austria, Lithuania and Portugal, instead, withdraw relatively little compared to the size of their wallets. Withdrawal frequency is also highly heterogeneous: in Finland and France, people make a withdrawal every three weeks, while Irish and Portuguese consumers withdraw almost once a week on average. In Slovakia, agents go to withdraw money when they still have 40 euros on hand (half of their usual cash balances); in Portugal, consumers visit ATMs when they have less than 10 euros (one fifth on their usual balances). Cashless acceptance is highly heterogeneous as well: it ranges from almost perfect (as in Finland) to very low (Austria,

FIGURE 20: Fit of calibrated models for Finland, Ireland, Italy and Portugal.



Note: Model moments are obtain through simulations based on optimal policy functions, by randomly drawing a stochastic stream of transaction sizes for a panel of agents. *Source of data: ECB SPACE (2019) Data.*

Italy and especially Slovakia). In some countries (Estonia, Finland, France) agents settle the vast majority of their expenditures using cards; in other (Italy, Greece, Spain) cash is still used to carry out the majority of payments in terms of value.

For the calibration exercise, I decide to pick four countries which are extremely different in terms of these moments; they have unique features that make them interesting case studies. The countries I choose are Finland (highest card acceptance and share of cashless payments in Europe), Ireland (high cash holdings and extremely big withdrawals despite widespread card acceptance), Italy (among the countries with the lowest card acceptance and cashless usage in Europe, paired with big holdings and withdrawals) and Portugal (low cash holdings and small withdrawals despite not-so-high cashless acceptance).

5.2 Calibration results

For all four countries, it is easy to recover the level of card acceptance ϕ , as well as the scale and location parameters of the log-normal transaction size distribution (μ_s and σ_s) directly from the data. There are five parameters of the model which I have to calibrate qualitatively: the discount factor β , the opportunity cost of cash R , the fixed

cost κ of using cashless, as well as fixed costs for voluntary and forced withdrawals, respectively η_l and η_h . The discount factor is set to $\beta = 0.995$ for all countries. I also set $\eta_h = 1$ as a normalization, to focus on parameters which are more meaningful from the payment choices and cash management perspective. The remaining three parameters are chosen to match five of the six moments displayed in Table 3, with the obvious exception of ϕ which can be set equal to its data counterpart. To compute moments implied by each parametrization, I solve the model numerically and I simulate a panel of individuals managing their cash holdings and deciding on how to pay based on the model's policy function. I then compute each of the model using data on the last period in the data for each agent (the day of the payment diary, in the simulated world).

In Figure 20 I display a comparison between simulated moments obtained from calibrated versions of the model and actual moments taken from SPACE data. It is possible to see that the model is able to replicate most qualitative features of the data for all countries. For instance, it can capture extremely high cash holdings and large withdrawals in the Irish setting, despite high levels of acceptance; it can also match low cash holdings and small withdrawals by Portuguese households in a low-acceptance environment. There are, of course, some shortcomings that could be due to the fact that a proper estimation through simulated method of moments was not performed. For instance, the model has some problems in fitting small levels of $\underline{M}$ observed in the data, and also to match withdrawal frequencies²⁰. This last problem also could be due to the fact that in my model I do not have multiple payments each day, so it's not easy to map it to my data structure in which agents make several payments (and more payments they make, more likely it is that they will withdraw). A version of the model with multiple payments every day would probably allow for sharper results, and it is needed to match data coming from a payment diary with the SUCH/SPACE structure.

In Table 4 I display the calibrated parameters that I picked in order to try and match the chosen moments. The differences in parameters among countries should not be taken from a quantitative point of view but just as an indication of their qualitative differences, as a formal estimation exercise has not been laid down. Still, the heterogeneity in parameters needed to reproduce the observed differences in payment and cash management behavior are noteworthy. In order to obtain high cash holdings and big withdrawals in Ireland, for instance, one needs big withdrawal costs and a pretty high fixed cost of using the card. Similar parameters must be set to obtain simulated moments close to the Italian ones. Portuguese and Finnish people seem to have different tastes (a much lower κ). In the Portuguese case, moreover, the calibrated η_l must

²⁰Part of the lack of fit is probably due to the lack of a formal estimation procedure at this stage. Fitting the model for each country by finding the combination of parameters which minimizes some weighted distance between simulated and actual moments, through a simple SMM procedure, is surely possible and easy to implement. This exercise will be present in the next version of the paper. Despite this, I want to stress that estimating parameters of the model is not the focus of this work; estimation is a big deal in this setting where choice sets are endogenous and there is a double discrete choice, and it is left to another paper.

TABLE 4: Calibrated parameters for Finland, Ireland, Italy and Portugal.

<i>Country</i>	η_l	R	κ
Finland	0.0357	0.00012	0.00357
Ireland	0.133	0.0002	0.0108
Italy	0.112	0.0003	0.0161
Portugal	0.005	0.000437	0.001

Note: The discount factor is set equal to $\beta = 0.995$ for all countries. The cost of running out of cash being forced to withdraw is set to $\eta_h = 1$ for all countries. Other parameters are set to match data moments as explained in the main text.

be extremely low to obtain the big and frequent withdrawals observed in the data. A more formal estimation exercise, employing rigorous methods from the dynamic discrete choice literature to a quantitative version of this model (as I propose in the next Subsection) could enable researchers to achieve parameter identification (under some assumptions), and to shed light on the quantitative differences in tastes (κ), technology (η_l and η_h) and environmental variables (R) that cause such big differences across European countries.

5.3 Towards a quantitative model

The model presented in this paper is extremely simple and it does not allow for quantitative analysis. First, decision rules are deterministic: depending on parameter values, for each (m, s) pair there is only one optimal action. This imposes restrictions which are not likely to hold in the data, and indeed they don't, as I showed in [Figure 16](#).

Deterministic payment choice policy functions that only depend on observables would create a zero likelihood problem in a proper estimation exercise. A proper quantitative model would include choice-specific utility instead of the unique parameter κ to proxy for the relative costs of paying cash or cashless at POS. Utility could depend on s , for instance, so as to incorporate supply-side factors. Moreover, one would need some unobserved component, such as random utility shocks to put nonzero probabilities on each choice and to avoid the zero likelihood problem.

In my setting, withdrawal policy functions are also deterministic, depending only on m . A way to avoid this would be to allow for random fixed costs, as proposed by [Alvarez and Lippi \(2017\)](#): in the data, agents do not only receive money by withdrawing at ATMs; sometimes they just get it from a cash jar at home or as cashback when shopping at a store. In fact, the data shows that around 15% of upward cash adjustments have these sources. Allowing for random variation in η_l and η_h (and estimating a distribution of fixed costs, instead of a unique value) would allow for more flexible adjustment rules. Moreover, in the model there is a unique target level of cash balances:

after a withdrawal, $m = m^*$ in any case. In reality, however, post-withdrawal balances have a non-degenerate distribution, even at the individual level. A way to obtain a non-degenerate distribution would be to allow individuals to receive a signal about the size of the incoming payment when making their withdrawal choices: in this way, they would decide to withdraw more (and more often) when there is a big incoming transaction, then when they think a small purchase will be the next one, mirroring the anticipatory behavior we see in [Figure 4](#).

Another shortcoming of the model is that agents never deposit cash, as they cannot have more cash than what they desire. Albeit there is no specific information on deposits in SUCH/SPACE data, future payment diaries could incorporate this information, allowing for the estimation of more flexible adjustment rules. In an extended version of this model, one could allow for exogenous *inflows* of cash: in SUCH/SPACE data, I observe that around 25% of cash is obtained from family, friends, colleagues or as cash income. Including inflows of cash would however make the problem a two-sided inventory problem, and the associated policy function would be much trickier than those studied here, and maybe close to the standard adjustment rules of two-sided models like [Miller and Orr \(1966\)](#).

The inclusion of these feature would allow one to take seriously the estimation procedure of parameters using tools from the dynamic discrete choice literature ([Rust \(1987\)](#), [Bajari, Benkard, and Levin \(2007\)](#)). An estimated, comprehensive model of payment choice and cash management would be a key tool to study the consequences and welfare implications of measures limiting cash usage or subsidizing card ownership. Moreover, it could be used to study the reasons behind the resilience of cash as a means of payment despite the existence of cheap substitutes, disentangling the effect of preferences for cash from that of non-universal acceptance of payment cards, and ultimately establishing the conditions under which the transition to a cashless society could be achieved.

6 Conclusion

High-quality, transaction level data on payment method decisions of Euro Area consumers shows that existing theoretical models of payment choice fail to capture key facts of observed behavior. In this paper, I exploit evidence from European payment diaries to argue that a model of payment choices needs to: i) jointly study means of payment and cash management decisions; ii) take into account the fact that consumers' decisions on how to pay are affected *both* by the size of each transaction and by the level of cash on hand, and not only by one among these factors, as previously assumed in the literature; iii) avoid abstracting from imperfect acceptance of cashless methods. I present a novel framework to study cash management and payment choices, showing its theoretical properties and empirically verifying some of its predictions. The model

account for observed choices that previous theories could not explain: in particular, it is the first to rationalize the fact that agents sometimes pay with cash even when they have enough cash holdings, and even if they state they prefer to use cash *ex-ante*: a key result is that this kind of behavior can arise as long as cashless payments are an effective cash management tool, i.e., they are “*cheaper*” than ATM withdrawals. I then show that a calibrated version of the model can match cross-country heterogeneity in payment and cash management choices. I conclude by arguing that the model presented in the paper could be enhanced by the inclusion of features that would make it suitable for quantitative analysis. Future research on payment choices could make use of such a quantitative model to study the welfare effects of policies aimed at increasing cashless usage, such as limits to cash holdings or to cash payments, and to analyze the reasons behind the resilience of cash in modern advanced economies despite advancements in payment technologies.

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A Appendix

A.1 Data cleaning and payment choice sets

The raw version of the data made available by the ECB has a major shortcoming: participants are not asked to report the level of their cash holdings in the moments in which they perform transactions. The rationale behind this choice is not easy to understand, since cash holdings were included among the possible factors that shape the decision on how to pay in the questionnaire administered in the same survey.²¹ At a first glance, this lack of information may seem no big deal: given that agents reported their money balances at the start of the day, using the data on withdrawals and payments cash holdings should be straightforward to pin down at each point in time. Two issues arise nonetheless: first, a fraction of agents failed to report the timing of cash adjustments performed (in SPACE data no one did, as respondents were not asked); second, participants were not explicitly asked about possible cash deposits.

To minimize the loss of data related to these shortcomings, I adopted the following strategy. Concerning the first problem, I was able to recover the timing of cash replenishments for a share of the agents which failed to report it: using the fact that I can observe which payment method they employed in each transaction, I can pin down the timing of adjustments in all situations in which an adjustment is needed to explain a purchase (i.e., in which an agent purchased in cash a good worth more than her *unadjusted* cash holdings). All the individuals for which the timing of replenishments couldn't be pinned down were excluded from the data. The second issue can also be tackled exploiting the fact that agents reported their cash holdings at the end of the day. For all the agents for which we are able to track cash holdings, we can compute our predicted cash balances at the end of the day *assuming there were no deposits*. If the two figures differ, then an unreported deposit or replenishment must have taken place during the day, and we are therefore not sure that cash on end at time of each transaction is corresponding to the actual one. These observations were excluded as well, and so were the ones for which computed cash balances were negative at some point. Finally, all the observations for Malta and Cyprus were excluded as I found systematic differences between the reported cash at the end of the day and the one I computed²². The final sample I exploit for the analysis contains information on 176,593 transactions carried out by 100,471 individuals²³. For a subsample I have access to a larger amount

²¹The absence of cash holdings could be due to the fact that previous literature (with some exceptions) did not stress their impact on payment decisions, focusing only on the size of the transaction. Further evidence about this is provided in the questionnaire, in which respondents were asked “*When shopping in shops, what is the amount below which you would typically pay with cash instead of other payment methods?*”. This question implicitly assumes that payment decisions are monotonic in transaction sizes, thereby ignoring the impact of cash on hand on choices.

²²This does not happen with other countries. Maybe a coding error in the data provided by ECB?

²³A share of participants actually did not carry out any payment in the day of analysis. The subsample of actual payers is composed by 75,187 individuals.

of information, since these participants also filled in the questionnaire. The variables I have access to in the cleaned version of the data can be divided in three families. First, I have access to *demographic data* on all participants. Second, I can observe a substantial amount of *transaction-level data* for all transactions performed. Third, for a subsample of participants (which includes a fraction of agents interviewed in the SUCH survey and all agents interviewed in the SPACE survey), *questionnaire data* are also available.

Payment choice set. A key aspect of the data is that it allows, for a large share of transactions, to perfectly observe the payment choice set of individuals, i.e., if they could use cash, cashless methods or both. First, tracking cash holdings at the moment of payments makes it possible to compare them with the associated transaction size and thereby to establish (for all 100,157 payments) if it was possible to carry out the transaction using cash: clearly, it is not possible to pay with cash if the transaction size is larger than current cash holdings. Formally, let m_{it} be cash holdings of individual i at time t and s_{it} be the transaction size she faces. Our dummy for the possibility to use cash $CashPoss_{it}$ is constructed by

$$CashPoss_{it} = \begin{cases} 1 & \text{if } s_{it} \leq m_{it} \\ 0 & \text{if } s_{it} > m_{it} \end{cases}$$

Second, combining information on card acceptance by merchants provided by respondents, payment methods employed and questionnaire answers, it has been possible to determine for a large share of payments if cashless payments were really an option for agents. There are two situations in which we are sure that cashless payments were an available option. The first situation, of course, is the one in which agents *did* use cashless methods to settle the transaction. The second situation is identified when two conditions are met: first, cashless payments must be accepted as a way to carry out the transaction (a condition I could check for all payments); second, the agent must have access to a payment card²⁴ (which I could check for a large fraction of payments²⁵). Formally, let $CardOwn_i$ be a dummy equal to one if individual i owns a payment card (debit or credit) and to zero if she doesn't. Let $CashlessOwn_i$ be equal to one if individual i has access to at least a cashless payment method and equal to zero otherwise. Let $CashlessAcc_{it}$ be a dummy equal to one if for the payment faced at time t by individual i cashless payments are accepted. Let $Cash_{it}$ be a dummy equal to one if the payment method used by i at time t was effectively cash, and equal to zero otherwise.

²⁴Since I don't know exactly which payment methods are accepted at any given transaction, but I just know if any cashless payments are accepted, I assume that an agent *could* have made a cashless payment at a store in which cashless instruments are accepted *if and only if* he/she has access either to a debit or to a credit card or both. The reason behind this is that credit and debit cards are always accepted in stores that accept some cashless payment method, while (for instance) cheques are not. Thereby, saying that an agent which has access to cheques (or credit transfers) could have paid cashless at a store in which cashless methods are accepted would be very risky, while for agents that own either a credit or a debit card it seems reasonably safe.

²⁵This are all the payments made by respondents to the questionnaire, for whom we have detailed information concerning access to cashless instruments, plus all the payments made by agents which performed at least a card payment.

Our dummy for the possibility to use cashless methods $CashlessPoss_{it}$ is constructed in the following way.

$$CashlessPoss_{it} = \begin{cases} 1 & \text{if } Cash_{it} = 0 \vee (CardOwn_i = 1 \wedge CashlessAcc_{it} = 1), \\ 0 & \text{if } CashlessAcc_{it} = 0 \vee CashlessOwn_i = 0, \\ \cdot & \text{otherwise,} \end{cases}$$

where $\cdot$ represent a missing value. For these transactions I can exactly pin down the payment choice set of individuals, and I'm thereby able to distinguish between *forced* (only cash or only payment card available) and *voluntary* (both available) payment choices. Formally, let Vol_{it} be a dummy variable equal to one when both payment options (cash and cashless) were available for individual i at time t and to zero when only one was available, which is constructed in the following way.

$$Vol_{it} = \begin{cases} 1 & \text{if } CashPos_{it} = 1 \wedge CashlessPos_{it} = 1 \\ 0 & \text{if } CashPos_{it} = 0 \vee CashlessPos_{it} = 0 \\ \cdot & \text{if } CashlessPos_{it} = \cdot \end{cases}$$

A.2 Proofs

Proof of Proposition 2

First, let's prove that $\exists \tilde{m}$ that satisfies (6). By the intermediate value theorem, given that V is continuous, that $V(\bar{m}) = V^* + \eta_l$ and that $V(m^*) = V^*$, $V(m)$ will take all the values in $(V^*, V^* + \eta_l)$. If $0 < \kappa < \eta_l/\beta$, then $V^* + \eta_l - \frac{\kappa}{\beta} \in (V^*, V^* + \eta_l)$.

We now prove points 1. and 2. of the Proposition.

1. By contradiction. Suppose that $\exists s$ such that $p(m, s) = 1$ for some $m \leq \tilde{m}$, where $\tilde{m}$ is given by (6). This would mean that, for some $m \leq \tilde{m}$, and for some s ,

$$\kappa + \beta V(m) < \beta V(m - s) \leq \beta (V^* + \eta_l),$$

since $V(m) < V^* + \eta_l$ for all m . This implies that for some $m \leq \tilde{m}$,

$$\beta V(m) < \beta (V^* + \eta_l) - \kappa = \beta V(\tilde{m}),$$

which is impossible for $m \leq \tilde{m}$ (it cannot hold for $m = \tilde{m}$ and it contradicts the fact that V is non-increasing in $[0, m^*]$ for $m < \tilde{m}$). Notice that for card usage to be optimal at $\tilde{m}$, one would need

$$\kappa + \beta V(\tilde{m}) = \beta (V^* + \eta_l) < \beta V(\tilde{m} - s),$$

which clearly cannot hold.

2. Let $m > \tilde{m}$. First, I show that for all such m the set $S(m)$ is nonempty, i.e., that

there exist at least one payment size for each $m > \tilde{m}$ such that it is optimal to pay cashless. Simply pick $m = \tilde{m} + \varepsilon$, for some $\varepsilon > 0$. We proceed by contradiction. Assume that for all s it is optimal to pay using cash, i.e., that

$$\kappa + \beta V(\tilde{m} + \varepsilon) > \beta V(\tilde{m} + \varepsilon - s).$$

Pick $s = \tilde{m} + \varepsilon - \bar{m}$. Then, I get

$$\kappa + \beta V(\tilde{m} + \varepsilon) > \beta V(\bar{m}) = \beta(V^* + \eta_l).$$

Rearranging,

$$\beta V(\tilde{m} + \varepsilon) > \beta(V^* + \eta_l) - \kappa,$$

that can be rewritten as

$$\beta(\tilde{m} + \varepsilon) > \beta V(\tilde{m}),$$

which clearly contradicts the fact that V is non-increasing in $[0, m^*]$. Call $\bar{s}(m) = m - \bar{m}$, the transaction size that decreases current money holdings to the trigger value $\bar{m}$. We just showed that $\bar{s}(m) \in S(m)$ for all $m > \tilde{m}$.

We want to show that $\exists \underline{s}(m)$ such that $p(s, m) = 1$ for all $\underline{s}(\tilde{m} + \varepsilon) < s \leq m$. For $m = \tilde{m} + \varepsilon$ and every $\varepsilon > 0$, $\underline{s}(m)$ is given by

$$\kappa + \beta V(\tilde{m} + \varepsilon) = \beta V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)).$$

Rearranging, we obtain

$$V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)) = V(\tilde{m} + \varepsilon) + \frac{\kappa}{\beta}.$$

Notice that $V(\tilde{m} + \varepsilon) + \frac{\kappa}{\beta} \leq V(\tilde{m}) + \frac{\kappa}{\beta} = V^* + \eta_l$. Again by the Intermediate Value Theorem, it certainly exists $\underline{s}(\tilde{m} + \varepsilon)$ that satisfies this Equation. It is clearly the case that for any $s > \underline{s}(\tilde{m} + \varepsilon)$,

$$V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)) > V(\tilde{m} + \varepsilon) + \frac{\kappa}{\beta},$$

so that it is optimal to use cashless methods for all $s \in (\underline{s}(m), m)$. Moreover, applying the Implicit Function Theorem to the Equation

$$J(m, \underline{s}) = V(m - \underline{s}) - V(m) - \frac{\kappa}{\beta} = 0,$$

we obtain

$$\frac{\partial \underline{s}(m)}{\partial m} = -\frac{\frac{\partial J}{\partial m}}{\frac{\partial J}{\partial \underline{s}}} = -\frac{V'(m - \underline{s}) - V'(m)}{-V'(m - \underline{s})} = -\frac{< 0}{> 0} > 0,$$

where the numerator is negative by convexity of V and the denominator is certainly nonzero because $V'(m) = 0$ only if $m = m^*$. Thus, $\underline{s}'(m) > 0$, as desired.

A.3 Discontinuous and non- η_l -convex $EV(m)$

In this Subsection, I show that it is not necessarily true that i) $EV(m)$ is continuous, and ii) $EV(m)$ is η_l convex. I do this using two examples, which are inspired by [Bar-Ilan \(1990\)](#), who uses similar examples to show the sub-optimality of (s, S) rules in a two-sided inventory problem.

Example 1 ($EV(m)$ is not continuous). I pick all parameters from my baseline parameter set (the one for which I plotted numerical results). Then I pick the following density functions for transaction sizes.

$$f(s) = \begin{cases} 1 & \text{if } s = 25, \\ 0 & \text{if } s \neq 25. \end{cases}$$

The numerical solution for this problem delivers value functions of the form plotted in [Figure 21a](#). We see that there are discontinuities at $s = 25, s = 50, s = 75$, etc. This is clearly due to the fact that the value of the problem changes suddenly at these thresholds. For instance, think about having $s = 24.99$ vs $s = 25$: the opportunity cost of cash is almost the same, but the agent will be able to pay for the next payment using cash; instead of using the card (if possible) or being forced to withdraw (if she runs into a store where cards are not accepted). A similar reasoning applies to the other discontinuities, the only difference being that the agent is able to afford two, three, four payments instead of one, before withdrawing again. Despite $EV(m)$ being discontinuous, an $(\bar{m}, m^*)$ -rule is still optimal in this case: agents start with $m = m^* = 125$ after a withdrawal; 5 cash payments later, they run out of cash, withdraw and the cycle repeats. Notice that when $m = 25$ agents prefer to wait, in order to exploit the cash they already have on hand in their next payment. As the payment size is deterministic ($s = 25$), it's clear that agents will never have $m \in (0, 25)$: they will therefore withdraw only when $m = 0$.

Example 2 ($EV(m)$ is not η_l -convex). Also in this example parameters are set at their baseline values. The density function for transaction sizes is given by

$$f(s) = \begin{cases} 0.01 & \text{if } s = 5, \\ 0.39 & \text{if } s = 100, \\ 0.6 & \text{if } s = 200. \end{cases}$$

In this case, the value function $EV(m)$ is neither continuous nor η_l -convex. The failure of η_l -convexity can be trivially established by looking at the definition of an η_l -convex function. This implies that simple $(\bar{m}, m^*)$ -rules are no longer optimal. When withdrawing, agents find still convenient to set their money balances at a level $m^* \simeq 100$.

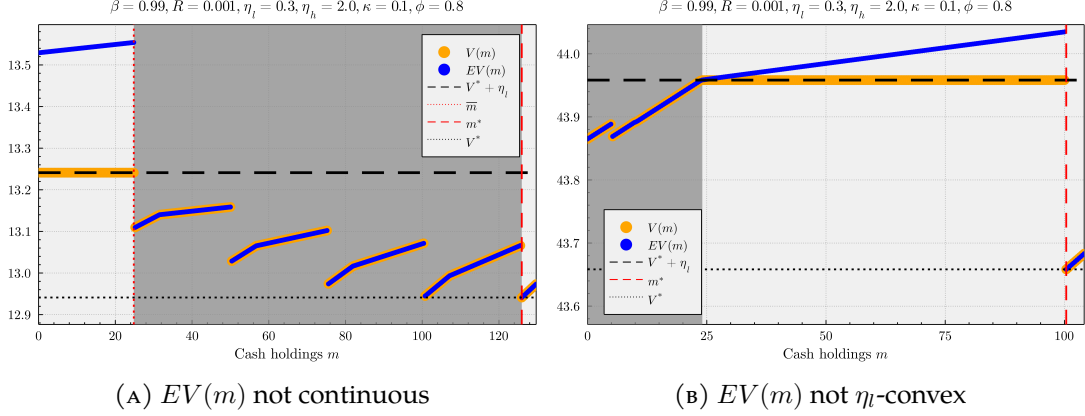


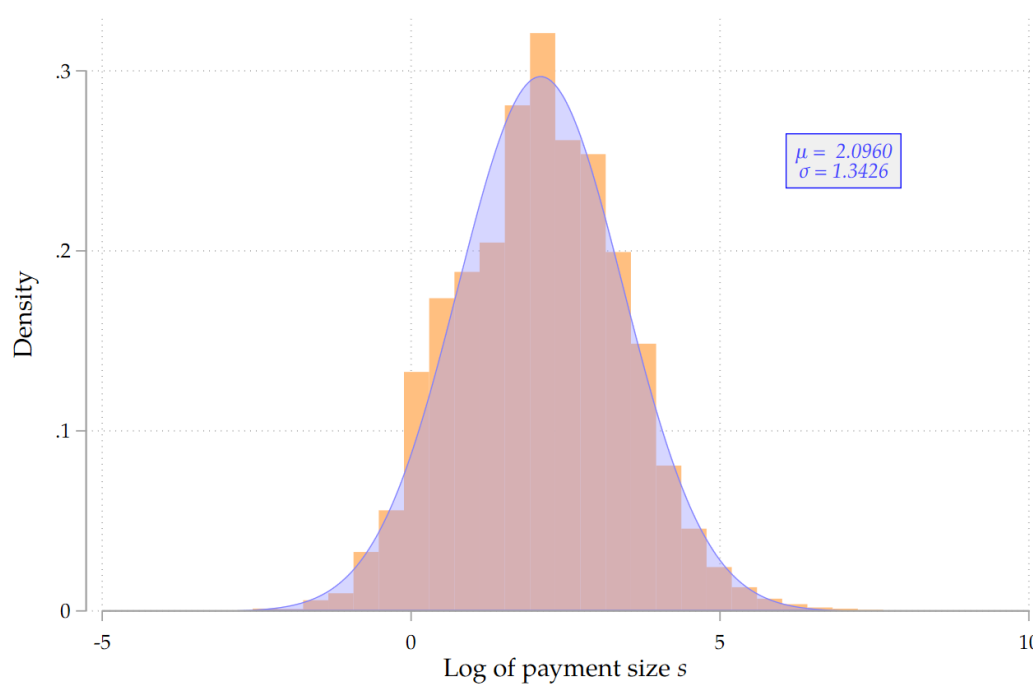
FIGURE 21: Cash management policies and value functions when density $f(\cdot)$ is degenerate.

However, there is no inaction region close to the optimal value: for levels of cash balances lower than m^* it is immediately optimal to reset cash holdings to the target level. Notice now that for $m < 100$, there is almost no precautionary value of holding cash (except for the ability to meet the very infrequent expenditures of size $s = 5$). Therefore, the value of the problem is increasing in m for $m < 100$: all amounts have more or less the same precautionary value, but bigger amounts entail higher opportunity costs. For m smaller than some threshold opportunity costs become so low that it is convenient for agents not to withdraw money, as the benefit they would get from withdrawing (being able to meet a transaction of size $s = 100$) is relatively small, and they still would not be able to meet the most frequent transaction of size $s = 200$. Therefore, the inaction region of this problem will be $[0, \underline{m}]$ for some $\underline{m} \simeq 25$. This is a completely different rule of behavior with respect to standard $(\bar{m}, m^*)$ -rules. This shows that these rules are not always optimal in this kind of problem due to the introduction of a payment method choice.

A.4 The transaction size distribution $f(s)$

The transaction size distribution is a key object in the model, as it has a first order impact on the shape of the functions $V(m)$ and $EV(m)$, as shown in [Appendix A.3](#). [Boeschoten \(1992\)](#) has noted that the distribution of transaction sizes is approximately lognormal. I investigate whether this is an appropriate assumption using all transactions in SUCH and SPACE data (more than 150,000 payments). As [Figure 22](#) shows, it turns out the $\log(s) \sim N(\mu, \sigma^2)$, therefore $s \sim LN(\mu, \sigma^2)$ appears to be an appropriate assumption. This is the specification we exploit to solve the model and obtain Figures and calibration results.

FIGURE 22: Voluntary cashless usage by cash lovers and implied cash remaining.



Note: This graph is based all transactions in the data (155,546 payments). It displays the histogram of log payment sizes with an overlaid normal fit; estimated parameters of the normal distribution are also displayed. *Source: ECB SUCH (2016) and SPACE (2019) Data.*